GL BAJAJ
Institute of Technology & Management
FIND YOUR SPARK

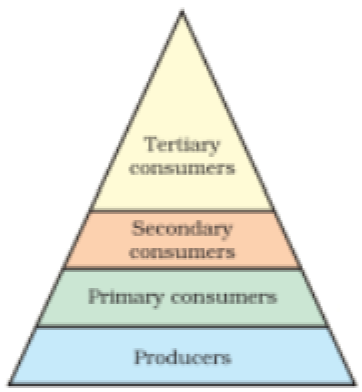
[redacted]

Course Outcomes:
Following are the course outcomes of the subject: -

CO's Codes	COURSE OUTCOMES	Blooms Level
BAS104.1	Gain in depth knowledge on natural processes that sustain life and govern economy.	**K2**
BAS104.2	Estimate and predict the consequences of human actions on the web of life, global economy and quality of human life.	**K3**
BAS104.3	Develop critical thinking for shaping strategies (scientific, social, economic and legal) for environment protection and conservation of biodiversity, social equity and sustainable development.	**K4**
BAS104.4	Acquire values and attitudes towards understanding complex environmental economic social challenges and participate actively in solving current environmental problems and preventing the future ones.	**K3**
BAS105.5	Adopt sustainability as a practice in life, society and industry.	**K3**

UNIT 1

S.No.	Very Short Questions	Marks
Q 1	**What is the role of atmosphere?** The atmosphere is a protective layer that safeguards the earth from the harsh conditions of the solar system. The air is a poor conductor of heat. During the day, it prevents a sudden increase in the temperature of the earth and prevents the heat to escape to outer space during the night. It keeps the average temperature of the earth steady. **Function of the Atmosphere** The atmosphere protects the earth from the radiation and cosmic rays coming from outer space. The ozone layer in the earth's atmosphere protects the earth from harmful ultraviolet radiations. The dense layers of molecular gases absorb the cosmic rays, gamma rays, and X-rays thereby blocking most of the harmful effects. Tons of space debris strike the earth every day. It is the molecules present in the atmosphere that destroy them before they reach the ground. The water vapour condenses and falls as rain, thereby, providing the earth with a life-sustaining resource. If there was no atmosphere, the water would have boiled away into space or would have remained frozen. **Importance of the Atmosphere** The earth's atmosphere protects the inhabitants by absorbing harmful solar rays and maintaining a steady temperature. It wards off many dangers of space thereby making life possible on earth. It is thus an important factor of climatic control and a life-sustaining source on earth.	**2**
Q 2	**Define Biosphere**. The biosphere is a narrow zone on the surface of the earth where soil, water, and air combine to sustain life. Life can only occur in this zone. From fungi and bacteria to large animals, there are several different types of life. The biosphere is characterized as an area that contains all living organisms and the products of	**2**

	their activities. As a result, it plays a critical role in the maintenance of ecosystems, i.e., the existence of species and their reciprocal interactions. And the biosphere is critical for climate regulation.	
Q 3	**The flow of energy is unidirectional and continuous in an ecosystem. Explain.** The energy flow in an ecosystem occurs from the lower trophic to the higher trophic level unidirectionally. Organisms in an ecosystem can survive due to the energy flow. Sunlight is the major primary source of energy in an ecosystem.We only receive less than 50 percent of the sun's effective radiation on earth. The flow of energy in an ecosystem follows the 10% rule, meaning only 10% of the energy is transferred to the successive trophic level and the rest is lost in the atmosphere. The energy is produced by the autotrophs, as they have photosynthetic pigments to harness the sunlight into chemical energy via photosynthesis. Most of the energy is lost as it moves to the successive trophic level. Hence, the amount of energy available in an ecosystem gets drastically less. The top consumers have the least amount of energy and the primary producers have the most energy, as the energy flows from the producers to the consumers. Thus, all these reasons indicate how the flow of energy is unidirectional in an ecosystem. 	**2**
Q 4	**Name two effects of mining on the people.** The most well-known adverse effects include air and water pollution, land clearing, soil erosion, destruction of biodiversity, health-related issues, and reduced community cohesion in mining areas	**2**
Q 5	**What is the Environmental Lapse Rate?** The rate at which the air temperature changes with height in the atmosphere surrounding a cloud or a rising parcel of air. The overall average rate is a decrease of about 6.5 °C/km, but the rate varies greatly in different regions of the world, in different airstreams, and at different seasons of the year.	**2**
Q 6	**Define Bioremediation.** Bioremediation is a biotechnical process, which abates or cleans up contamination. It is a type of waste management technique which involves the use of organisms to remove or utilize the pollutants from a polluted area. Bioremediation is the use of microbes to clean up contaminated soil and groundwater. Microbes are very small organisms, such as bacteria, that live naturally in the environment. Bioremediation stimulates the growth of certain microbes that use contaminants as a source of food and energy.	**2**
Q 7	**What is Agenda 21?** It is the declaration signed by world leaders in 1992 at the United Nation's Conference on Environment and Development which took place at Rio de Janeiro (Brazil). It aims at achieving global sustainable development. It is an agenda to combat environmental damage, poverty, disease through global cooperation on common interests, mutual needs and shared responsibilities. Agenda 21 aims to protect the environment and natural resources for present and future generations. It calls for the conservation of biological diversity, the sustainable use of natural resources, and the prevention of pollution and degradation of the environment.	**2**
Q 8	**Write any 5 sustainable development goals.** • No poverty. • Zero hunger.	**2**

	• Good health and well - being. • Quality education. • Gender Equality. • Clean water and sanitation. • Affordable and clean energy.	
Q 1	**Describe the characteristic features, structure and function of pond ecosystem.** A pond ecosystem is a freshwater ecosystem that can either be temporary or permanent and consists of a wide variety of aquatic plants and animals interacting with each other and the surrounding aquatic conditions. **Abiotic Components of the Pond Ecosystem** Abiotic components are the non-living components of an ecosystem that matter for the aquatic species' survival. There are the following main abiotic components of a pond ecosystem: • Light: Light serves as a main abiotic component required for the photosynthetic activities of the phytoplankton. The littoral zone has the maximum light penetration, whereas the profound zone has the least light penetration. • Temperature: As the depth of the pond increases, the temperature of the water gradually decreases due to the gradual decrease in the light penetration. • Dissolved oxygen: The amount of dissolved oxygen is maximum in the shallow water and gradually decreases while moving from the surface to the depth of the pond. • Dissolved oxygen: The amount of dissolved oxygen is maximum in the shallow water and gradually decreases while moving from the surface to the depth of the pond. **Biotic Components of the Pond Ecosystem** Biotic components are living components. A wide variety of living components are found in the pond ecosystem can be discussed as follows • Producers: These include species of rooted, submerged, emerged, floating plants and algae. The most common filamentous algae found in ponds is Spirogyra. • Primary consumers: A large population of zooplanktons are the main primary consumers. Besides these, small herbivores such as snails, insects, small fishes, tadpoles, and larvae of aquatic animals are the primary consumers often found in the pond. • Secondary consumers: These include large animal species such as frogs, big fishes, water snakes, crabs, etc. The consumers of the highest order might include mammals like water shrews, water voles, herons, ducks, kingfishers, etc. • Decomposers: These include different types of bacteria and fungi that feed upon dead and decaying parts of the aquatic species. **Functions of Ecosystem** 1. It regulates the essential ecological processes, supports life systems and renders stability. 2. It is also responsible for the cycling of nutrients between biotic and abiotic components. 3. It maintains a balance among the various trophic levels in the ecosystem.	**7**

4. It cycles the minerals through the biosphere.
5. The abiotic components help in the synthesis of organic components that involve the exchange of energy.

Q 2 Define ecology and ecosystem. Describe the structures and functions of forest ecosystem. **7**

ECOLOGY Ecology is a branch of science, including human science, population, community, ecosystem and biosphere. Ecology is the study of organisms, the environment and how the organisms interact with each other and their environment.

ECOSYSTEM An ecosystem is the structural and functional unit of ecology. An ecosystem describes the relationship of the mass of living organisms, which belong to same or different communities, with each other and the surrounding environment. An ecosystem is a chain of interaction between organisms and their environment.

Forest Ecosystem: The terrestrial system in which living things such as trees, insects, animals, and people interact is referred to as a forest ecosystem. It is the smaller classification of the ecosystem as a whole, which is the biggest functional unit comprising all the geographical features and living organisms on Earth. There are many different kinds of forest ecosystems, and they are categorised according to the local climate, including the amount of rainfall and temperature.

Components of Forest Ecosystem

Biotic components

1. Producers: Producers can synthesise their own food by the photosynthesis process. All green plants are considered producers of the ecosystem as they convert sunlight into the chemical energy of food.
2. Primary Consumers: Since the consumers can not prepare their own food, they depend on producers. Herbivorous animals get their food by eating the producers (plants) directly. Examples of primary consumers are grasshoppers, deer, etc.
3. Secondary Consumers: Secondary consumers draw their food from primary consumers.
4. Decomposers: The decomposers of the forest ecosystem break down dead plants and animals, returning the nutrients to the soil so that they can be used by the producers. Apart from bacteria, ants and termites are important decomposers.

Abiotic components

These include basic inorganic & organic compounds present in the soil & atmosphere. In addition dead organic debris is also found littered in forests.

Functions of a Forest Ecosystem

Forest ecosystems play a quintessential role by balancing the climate of a particular environment. Arrays of trees help maintain te balance of carbon dioxide and oxygen in the atmosphere, as well as preventing soil erosion.

A forest ecosystem is a conglomeration that provides shelter to millions of plants, animals and micro-organisms and also to a variety of insects, birds, trees and humans. Ideally any natural woodland area comes under the heading of forest ecosystem, and is a suitable place for the survival of equipping biotic and abiotic components.

Q 3 **Explain about various objectives and scope of environmental studies.** **7**

Environmental studies describe the interrelationships among organisms, the environment and all the factors, which influence life on earth, including atmospheric conditions, food chains, the water cycle, etc. It is a basic science about our earth and its daily activities, and therefore, this science is important for one and all.

Scope of environmental studies

Environmental studies discipline has multiple and multilevel scopes. This study is important and necessary not only for children but also for everyone. The scopes are summarized as follows:

1. The study creates awareness among the people to know about various renewable and nonrenewable resources of the region. The endowment or potential, patterns of utilization and the balance of various resources available for future use in the state of a country are analysed in the study.
2. It provides the knowledge about ecological systems and cause and effect relationships. 3. It provides necessary information about biodiversity richness and the potential dangers to the species of plants, animals and microorganisms in the environment.
4. The study enables one to understand the causes and consequences due to natural and main induced disasters (flood, earthquake, landslide, cyclones etc.,) and pollutions and measures to minimize the effects.
5. It enables one to evaluate alternative responses to environmental issues before deciding an alternative course of action.
6. The study enables environmentally literate citizens (by knowing the environmental acts, rights, rules, legislations, etc.) to make appropriate judgments and decisions for the protection and improvement of the earth.
7. The study exposes the problems of over population, health, hygiene, etc. and the role of arts, science and technology in eliminating/ minimizing the evils from the society.
8. The study tries to identify and develop appropriate and indigenous eco-friendly skills and technologies to various environmental issues.
9. It teaches the citizens the need for sustainable utilization of resources as these resources are inherited from our ancestors to the younger generating without deteriorating their quality.
10. The study enables theoretical knowledge into practice and the multiple uses of environment.

IMPORTANCE

	World population is increasing at an alarming rate especially in developing countries. • The natural resources endowment in the earth is limited. • The methods and techniques of exploiting natural resources are advanced. • The resources are over-exploited and there is no foresight of leaving the resources to the future generations. • The unplanned exploitation of natural resources lead to pollution of all types and at all levels. • The pollution and degraded environment seriously affect the health of all living things on earth , including man. • The people should take a combined responsibility for the deteriorating environment and begin to take appropriate actions to space the earth. • Education and training are needed to save the biodiversity and species extinction. • The urban area, coupled with industries, is major sources of pollution. • The number and area extinct under protected area should be increased so that the wild life is protected at least in these sites. • The study enables the people to understand the complexities of the environment and need for the people to adapt appropriate activities and pursue sustainable development, which are harmonious with the environment. • The study motivates students to get involved in community action, and to participate in various environmental and management projects. • It is a high time to reorient educational systems and curricula towards these needs. • Environmental studies take a multidisciplinary approach to the study of human interactions with the natural environment. It integrates different approaches of the humanities , social sciences, biological sciences and physical sciences and applies these approaches to investigate environmental concerns.	
Q 4	**What do you mean by ecological pyramids? Enlist the various types, importance, and drawbacks of these pyramids.** **ECOLOGICAL PYRAMIDS** They are graphical representation of various ecological parameters at the successive trophic levels of the food chain with producers at base, top carnivores at apex and intermediate levels in between. The common parameters used in preparing ecological pyramids are numbers of individuals, biomass and energy at different trophic levels. **1) Pyramid of numbers** - In this type of ecological pyramid, the number of organisms in each trophic level is considered as a level in the pyramid. a) Upright- In most ecosystem, the number of producers are maximum thus producers can support fewer herbivores and herbivores can support still fewer carnivores and so on. Thus the number of top carnivore too small to support any other trophic levels and do not act as prey to any other organisms.E.g. Pond ecosystem, Grassland ecosystem. b) Inverted- In some cases at each successive trophic level, number of the organisms as higher than in preceeding one and the size decreases gradually at successive level. Thus, shape of pyramid may be inverted. E.g. Tree ecosystem **2) Pyramid of biomass-** In this particular type of ecological pyramid, each level takes into account the amount of biomass produced by each trophic level. a) Upright- Pyramid of biomass is always upright in terrestrial ecosystems.For upright pyramid, total biomass of plants in a specific	**7**

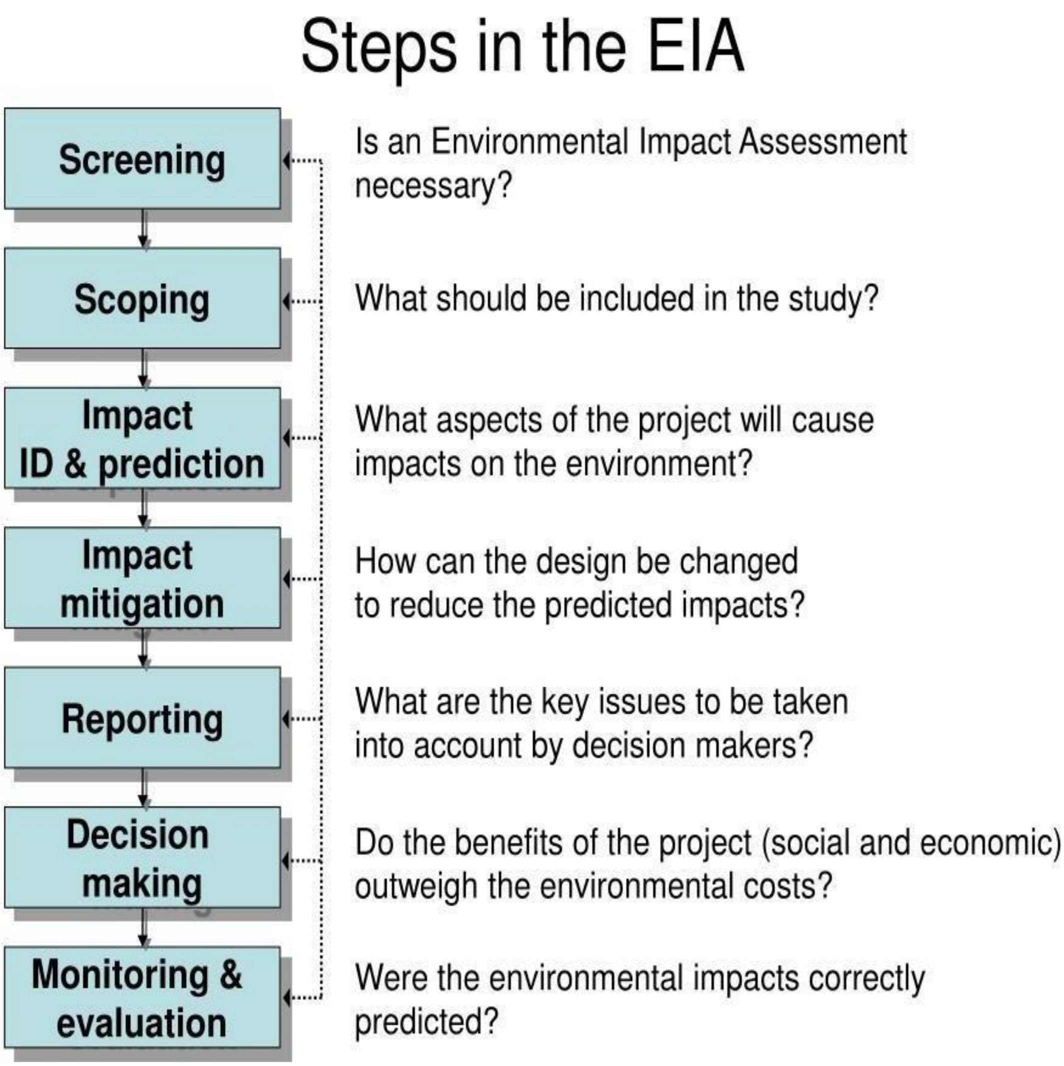
	area is more than that of herbivores and it gradually decreases at each successive trophic level.E.g. Tree ecosystem, Grassland ecosystem. b) Inverted- In an aquatic ecosystem, the pyramid of biomass is always inverted. Biomass of zooplankton is higher than that of phytoplankton as life span of former is longer and the latter multiply much faster though having shorter life span. A number of generations of phytoplankton may thus be consumed by single generation of zooplanktons. **3) Pyramid of energy**- The pyramid of energy is always upright because the flow of energy is always unidirectional from producer to consumer level. The energy content is maximum in producers. The energy decreases at each trophic level of food chain, as a part of energy is lost as heat. Importance of Ecological Pyramid **The importance of ecological pyramid can be explained in the following points:** 1. They show the feeding of different organisms in different ecosystems. 2. It shows the efficiency of energy transfer. 3. The condition of the ecosystem can be monitored, and any further damage can be prevented. **Limitations of Ecological pyramid** 1) It does not take into account the same species belonging to two or more trophic levels. E.g. Insectivorous plants 2) It assumes a simple food chain and does not accommodate a food web 3) Decomposers, microbes are not given any place in ecological pyramids.	
Q 5	**Discuss in detail about the term Environmental Impact Assessment (EIA). Describe the various steps involved in EIA process.** The EIA process is designed to ensure that proposed projects are carefully evaluated for their potential environmental, social, and economic impacts and that appropriate measures are taken to mitigate or avoid any negative impacts. The EIA process helps to promote sustainable development and protect the environment while ensuring that development projects are implemented responsibly and transparently. The stages of Environmental Impact Assessment involve Screening, Scoping, Preparation of the EIA Report, Application, and consultation, Decision Making, Post Decision. The stages of Environmental Impact Assessment are explained as follows: **Screening:** This is the initial stage of EIA, where the proposed project is evaluated to determine whether it requires a full EIA. Screening involves assessing the potential environmental impacts of the project based on factors such as the size, location, and nature of the project. **Scoping:** In this stage, the scope and objectives of the EIA are defined. Scoping involves identifying the project's potential environmental, social, and economic impacts and selecting the key issues that will be evaluated in the EIA. **Preparation of EIA Report**: This stage involves the preparation of a detailed report that describes the potential impacts of the project, evaluates the significance of these impacts, and recommends measures to mitigate or avoid any negative impacts. The EIA report is typically prepared by environmental experts based on data collected during field investigations,	**7**

stakeholder consultations, and other sources.

Making An Application and Consultation: In this stage of environmental impact assessment, the report is submitted to the regulatory authority and the application for project approval. The regulatory authority will then review the EIA report and may seek additional information or clarification. There is also often a public consultation process where stakeholders can provide feedback and comments on the project proposal and the EIA report.

Decision Making: Based on the EIA report, application, and public consultation process, the regulatory authority will decide whether to approve, reject, or request modifications to the project proposal. The decision may be based on environmental, social, economic, and other factors and compliance with relevant laws and regulations.

Post Decision: Once a decision has been made, the project can proceed with implementation, subject to any conditions or requirements specified in the approval. The regulatory authority will typically monitor the project to ensure compliance with the approval conditions and any environmental management plans that have been developed.



Q 6	**Mention the various anthropogenic activities that cause degradation of natural resources.**	**7**

Deforestation and Habitat Destruction

Deforestation, the relentless clearing of forests, stands as one of the most concerning human

	activities impacting the environment. As we chop down trees to make way for agriculture, infrastructure, and settlements, we disrupt delicate ecosystems and threaten countless plant and animal species. **Fossil Fuel Consumption and Greenhouse Gas Emissions** Our insatiable appetite for fossil fuels, such as coal, oil, and natural gas, drives much of our energy production. However, the combustion of these non-renewable resources releases copious amounts of greenhouse gasses (GHGs) into the atmosphere. **Industrialization and Air Pollution** The rapid rise of industrialization has undoubtedly fueled economic growth and technological advancements. However, it has also given rise to a significant environmental concern – air pollution. **Agricultural Practices and Soil Degradation** Modern agriculture has undoubtedly enabled us to feed a growing global population, but it has also taken a toll on our precious soil. Intensive farming practices, such as heavy pesticide and fertilizer use, monoculture, and overgrazing, have led to soil degradation. **Water Pollution and Contamination** Water pollution, a dire consequence of human activities, poses a significant threat to our planet's aquatic ecosystems and human health. From industrial discharges to agricultural runoff and improper waste disposal, various sources contribute to contaminating our water bodies. **Overfishing and Marine Ecosystem Depletion** Overfishing, driven by the increasing demand for seafood, poses a grave threat to marine biodiversity and the delicate balance of marine ecosystems. As fishing practices become more efficient and industrialized, many fish populations face depletion, pushing them to the brink of collapse. **Waste Generation and Management** The modern world's consumer-driven lifestyle has led to an alarming surge in waste generation, creating a pressing environmental challenge. From plastic packaging to electronic waste, the improper disposal of waste poses significant risks to our planet. **Urbanization and Habitat Fragmentation** The rapid expansion of cities and urban areas has reshaped landscapes and posed significant challenges for the environment. Urbanization leads to habitat fragmentation, as natural habitats are fragmented and isolated by roads, buildings, and infrastructure. **Consumerism and Resource Depletion** In a world driven by consumerism, the relentless pursuit of goods takes a toll on the Earth's finite resources. The ever-increasing demand for products like electronics, clothing, and food puts immense pressure on natural resources and ecosystems. **Climate Change and Global Impact** Climate change, driven primarily by human activities, is one of the most critical environmental challenges of our time. The burning of fossil fuels, deforestation, industrial emissions, and other human activities release greenhouse gasses into the atmosphere, leading to a warming planet.	
Q 7	**Explain the concept of sustainable development. What are the major obstacles in the path of sustainable development in India?** **SUSTAINABLE DEVLOPMENT** Sustainable development can be defined as an approach to the economic development of a country without compromising with the quality of the environment for future generations. The features of sustainable development are: 1. Minimising the release of greenhouse gases that are responsible for air pollution and global warming. 2. Using renewable sources of energy such as wind, solar energy etc to reduce the global carbon footprint. 3. Limiting the rate of consumption of natural resources so that it does not exceed the rate of its production. 4. Preservation of natural resources and protection of natural habitats of organisms. 5. Emphasis should be placed on promoting green architecture for building homes and offices. **Objectives**	**7**

The four objectives of sustainable development are :

1) Stable economic growth- The eradication of poverty and hunger to ensuring a healthy life.

2) Conservation of natural resources - Achieve universal access to basic services such as water, sanitation, and sustainable energy.

3) Social progress and equality - Reduce inequalities in the world, especially gender inequalities. Supporting the generation by the development of opportunities through inclusive education and decent work. Foster innovation and resilient infrastructures by creating communities and cities capable of producing and consuming sustainably.

4) Environmental protection - Caring for the environment by combating climate change and protecting oceans and terrestrial ecosystems.

Major obstacles in the path of sustainable development

There are numerous challenges to sustainable development in India. While many of these problems are caused due to insensitive use of natural resources, governmental responsibility is also trifling in solving the critical issues.

Deforestation The forest resources in India as well as around the world are on the verge of a higher depletion rate and are reaching alarming proportions. The individuals, corporations, government agencies etc., are responsible for this menace. In the name of developmental projects, the felling of trees is being carried out leaving behind the goals of sustainable development and human progress. The most serious problem of deforestation is the loss of biodiversity. The destruction of forests leads to not only the extinction of endangered animal species but also many plant varieties that have immense medicinal value. It is also responsible for global warming in a massive way.

Threat to Biodiversity The biodiversity of the earth is a crucial asset that needs to be conserved and utilised in a judicious manner. The fair and equitable sharing of these resources is a prerequisite for a good life. The massive habitat destruction, pollution of the land, water and soil has a drastic effect on the survival of biodiversity. The biological resources, due to injudicious use, are on the verge of extinction. Though the problem can be solved by applying serious restrictions on the excessive use, lack of collective will has greatly hampered the process of sustainable development at large.

Effects of Climate Change The drastic changes in the climatic variations resulted in poor health conditions of the human beings and earth resources. These have also spilled dire consequences on the social and environmental aspects of the society. The increasing temperature levels and the carbon emissions had severe effects like crop failures, increasing droughts, scarcity of food supply, contagious diseases, degradation of environment, increasing floods and so on. Lack of disaster management methods and systematic marginalisation of local communities in policy formulations have increased the vulnerabilities to the natural and man-made disasters.

Increasing Pollution Levels The degrading levels of air quality are widely recognised as a major factor of pollution, especially in urban areas. The sources of air pollution include industrial pollution, indoor and vehicular pollution. The pollution in urban areas is caused by the presence of a number of industries that emanate smoke and other chemical substances into the air. Added to this is the vehicular pollution that has been on an increase every year. The sale in the number of vehicles has been zooming at an unprecedented scale and leading to massive traffic congestions. This invariably has resulted in serious health hazards like asthma, respiratory problems, hearing impairment and so on.

Ground Water Depletion and Pollution While shortage of water continues to loom large, the inefficient use of water is an avoidable crisis, which otherwise can lead to imbalances in the water management methods. Apart from this, access to safe drinking water has also become a pertinent issue with major organic and bacterial pollutants being untreated. In many of the cities, untreated municipal waste/sewage is being discharged into the rivers.

Poor Health The developments in the health sector are confined basically to urban areas; moreover, the status of income too largely determines the access to it. Life expectancy levels, no doubt, have gone up but it is altogether important to note the high rates of infant mortality

Literacy Rate One of the basic and most important components of development is education. It is a critical invasive instrument for bringing about social, economic and political inclusion and a durable integration of people, particularly those ‘excluded’ from the mainstream of any society

Q 8	**State the chief characteristics of an ocean biome and discuss its structure and function.** The open ocean is the largest marine biome and covers over 70% of the Earth's surface. It is characterised by deep, dark waters and a lack of physical features such as coastlines and islands. The open ocean is home to a wide range of organisms, including plankton, fish, sea turtles, and whales. Like ponds and lakes, the ocean regions are separated into separate zones: intertidal, pelagic, abyssal, and benthic. All four zones have a great diversity of species. Some say that the ocean contains the richest diversity of species even though it contains fewer species than there are on land. There are five ocean biomes in the world – Atlantic Ocean, Pacific Ocean, Indian Ocean, Southern Ocean, and the Arctic Ocean. Marine ecosystems also provide other important services, associated with their regulatory and habitat functions, such as pollution control, storm protection, flood control, habitat for species, and shoreline stabilization.	**7**
Q 9	**Describe the various Goals of sustainable development by WHO.** The Sustainable Development Goals are a set of seventeen pointer targets that all the countries which are members of the UN agreed to work upon for the better future of the country. It is a group of 17 goals with 169 targets and 304 indicators, as proposed by the United Nation General Assembly's Open Working Group on Sustainable Development Goals to be achieved by 2030. Post negotiations, agenda titled "Transforming Our World: the 2030 agenda for Sustainable Development" was adopted at the United Nations Sustainable Development Summit. SDGs is the outcome of the Rio+20 conference (2012) held in Rio De Janerio and is a non-binding document. The 17 SDGs are: 1) No poverty 2) Zero Hunger , 3) Good health and well-being 4) Quality education 5) Gender equality 6) Clean water and sanitation 7) Affordable and clean energy 8) Decent work and economic growth 9) Industry, innovation and infrastructure 10) Reduced Inequality 11) Sustainable Cities and Communities 12) Responsible Consumption and Production 13) Climate Action 14) Life Below water 15) Life On Land 16) Peace, Justice, and Strong Institutions 17) Partnerships for the **Goals.**	**7**
Q 10	Write the strructure of an ecosystem? Explain the various components of an ecosystem structure. Discuss the functions of an ecosystem.	**7**

Ecology is the study of inter-relationship of organisms with physical as well as biotic environments. Ecology examines how living beings or organisms interact and relate with the environment.

An ecosystem is the structural and functional unit of ecology. An ecosystem describes the relationship of the mass of living organisms, which belong to same or different communities, with each other and the surrounding environment. An ecosystem is a chain of interaction between organisms and their environment.

1) **Biotic components** are the living things that have a direct or indirect influence on other organisms in an environment. For example plants, animals, and microorganisms and their waste materials.

Plants: Producers that convert sunlight into energy through photosynthesis and form the most terrestrial ecosystems.

- Animals: Consumers that rely on plants and other animals for food, contributing to ecological processes such as predation, herbivory, and decomposition.
- Microorganisms: Including bacteria, fungi, and protists, which play essential nutrient cycling, decomposition, and maintaining soil fertility.

Humans: As a species, humans profoundly influence the environment through activities such as agriculture, industry, and urbanization.

2) **Abiotic Components**:

Atmosphere: The layer of gases surrounding the Earth, including oxygen, nitrogen, carbon dioxide, and other trace gases. It plays a crucial role in regulating temperature, weather patterns, and the composition of the air. The atmosphere can be divided into layers based on its temperature. Atmosphere comprises of mixture of gases and is divided five layers i.e. troposphere, stratosphere, mesosphere, thermosphere and exosphere.

- Troposphere- This is the lowest part of the atmosphere. Troposphere means "region of mixing. It contains most of our weather – clouds, rain, snow. In this part of the atmosphere the temperature gets colder as the distance above the earth increases, by about 6.5°C per kilometer.

Stratosphere This extends upwards from the tropopause to about 50 km. It contains 90% of the total ozone in the atmosphere. The increase in temperature with height occurs because of absorption of ultraviolet (UV) radiation from the sun by this ozone.

Mesosphere-The region above the stratosphere is called the mesosphere. Here the temperature again decreases with height, reaching a minimum of about -90°C at the "mesopause".
The stratosphere and mesosphere together are sometimes referred to as the middle atmosphere. Thermosphere- The thermosphere lies above the mesopause, and is a region in which temperatures again increase with height. This temperature increase is caused by the absorption of energetic ultraviolet and X-Ray radiation

from the sun.

The Exosphere- The region above about 500 km is called the exosphere. The exosphere is the very edge of our atmosphere. This layer separates the rest of the atmosphere from outer space. It's about 6,200 miles (10,000 kilometers) thick. There is a lot of empty space in between. There is no air to breathe, and it's very cold.

2) Hydrosphere: Comprising all the water on Earth, including oceans, rivers, lakes, and groundwater. Water is essential for life and influences climate, weather patterns, and the distribution of organisms.

3) Geosphere: The solid part of the Earth, including rocks, minerals, soil, and landforms. It provides habitats for organisms and serves as a reservoir of nutrients and resources.

a. Climate: The long-term patterns of temperature, precipitation, humidity, wind, and other atmospheric conditions in a particular region. Climate influences the distribution of ecosystems and species.

Soil: A complex mixture of minerals, organic matter, water, air, and living organisms. Soil provides nutrients and support for plant growth and is essential for agriculture and terrestrial ecosystems.

c. Sunlight: The primary source of energy for life on Earth, driving photosynthesis and powering ecological processes.

Q 11 **Define food chain. Name and explain various types of food chains with suitable examples.** **7**

Food chain- It is a sequence of living organism which involves transfer of food energy from producers through a series of organisms with repeated eating and being eaten is called food chain.Each level or step in a food chain where transfer of energy takes place is called trophic level.

Types of food chain

1) Grazing food chain/ Predator food chain

2) Detritus food chain

3) Parasitic food chain

1) Grazing food chain- Consists of producers, consumers and decomposers. Source of energy for such food chain is sun.

Leaf → Caterpillar→ Chameleon → Mongoose →Snake

2) Detritus food chain-Begins with dead organic matter. It is made up of decomposers which are hetrotrophic organisms mainly fungi and bacteria. A common detritus food chain is given below.

Dead organic matter → Earthworm → Sparrow → Falcon

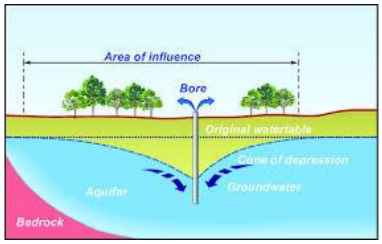
3) Parasitic food chain- Size of organism finally reduces at higher trophic level(parasite).

E.g. Tree → Herbivores birds→ Lices and bugs

Q 12 **The pyramid of energy always takes a true upright shape. Why?** **7**

Pyramid of energy is the only pyramid that can never be inverted and is always upright. This is because some amount of energy in the form of heat is always lost to the environment at every trophic level of the food chain. It is measured in terms of energy per area per time. The amount of energy transferred to the next level is measured by the transfer of productivity between the trophic levels and so is also known as the Pyramid of productivity. The energy lost between each trophic level is huge, which can be explained by the 10% law that states “each greater and succeeding trophic level receive only 10% of energy from the preceding trophic level”, it is also due to this reason that the number of levels in a pyramid of energy in a food chain is limited.

Unit 2

S.No.	Very Short Questions	Marks	CO's	BL
Q 1	***<u>What do you mean by Cone of depression?</u>*** A cone of depression is a conical-shaped depression in the water table that forms around a well that is being used to withdraw water. The cone of depression is also known as the cone of drawdown or cone of influence.  	**2**	**CO .2**	**K 3**
Q 2	What is the significance of D.O.? Dissolved oxygen (DO) is a key indicator of water quality and is essential for the survival of aquatic life. It's the amount of oxygen gas that's dissolved in water. • **Supports aquatic life**: DO is necessary for the growth and reproduction of aerobic aquatic life, such as fish and crustaceans. • **Indicates water quality**: DO levels can indicate how well a body of water can support aquatic life. • **Important for aquaculture**: Aquaculture companies need to maintain precise DO levels to raise healthy seafood.	**2**	**CO .2**	**K 3**
Q 3	Define OTEC. Ocean Thermal Energy conversion, (OTEC) is a technology that uses the ocean's temperature difference to generate electricity. It's a renewable energy source that can also produce desalinated water.	**2**	**CO .2**	**K 3**
Q 4	Name the various types of coal and its characteristics. Coal is classified into four main types, or ranks: anthracite, bituminous, subbituminous, and lignite. The ranking depends on the types and amounts of carbon the coal contains and on the amount of heat energy the coal can produce. Anthracite • **Characteristics**: Black, shiny, and hard, with up to 98% carbon • **Uses**: Used in industries that require high temperatures for combustion, such as steel foundries • **Formation**: Formed from bituminous coal under great pressure during the creation of mountain ranges Bituminous • **Characteristics**: Black, with 45–68% carbon • **Uses**: Used in power plant boilers and industrial boilers • **Formation**: Formed under high temperature and pressure conditions over 100–300 million years Sub-bituminous • **Characteristics**: Contains less carbon and more water than bituminous coal • **Uses**: Used for power generation • **Formation**: Most subbituminous coal in the United States is at least 100 million years old Lignite • **Characteristics**: Soft, with a range of colors from black to brown, and up to	**2**	**CO .2**	**K 2**

	40–42% moisture • **Uses**: Mainly used for power • **Formation**: The youngest form of coal			
Q 5	What are the various consequences of overuse of groundwater? Overuse of groundwater can have many negative effects, including: • **Lowered water table**: Pumping groundwater faster than it can be recharged lowers the water table, which can cause wells to dry up • **Land subsidence**: When the ground loses support from below, it can subside • **Reduced water in streams and lakes**: Groundwater contributes to the water in streams, lakes, and wetlands • **Increased pollution**: When groundwater is overpumped, it can reduce the flow of surface water from springs, which can increase pollution from fertilizers and nutrients • **Sea level rise**: As aquifers are depleted, less water is discharged into oceans and coastal areas, which can cause sea levels to rise • **Changes in aquatic ecosystems**: When groundwater is overpumped, it can change the balance of aquatic plant and animal species • **Increased costs**: When pumps are used to lift water, it requires more energy, which can increase costs • **Damaged structures**: Overuse of groundwater can damage roads, buildings, and other structures **Increased flooding**: Overuse of groundwater can lead to more frequent and severe flooding	**2**	**CO .2**	**K 3**
Q 6	What do you mean by biodiversity hotspot? Biodiversity hotspots are regions with a high level of biodiversity and a high percentage of endemic species. They are considered irreplaceable because they contain species that are found nowhere else on Earth.	**2**	**CO .2**	**K 2**
Q 7	Elaborate OTEC. Ocean thermal energy conversion	**2**	**CO .2**	**K 3**
Q 8	Write short note on Tidal energy. Tidal energy is a renewable energy source that uses the natural rise and fall of tides to generate electricity. It's a clean and reliable energy source that's more powerful than wind energy.	**2**	**CO .2**	**K 3**
	Long Questions			
Q 1	Explain water induced and water borne diseases along with their causes, symptoms, and preventive measures. **Water borne Diseases:** These are conditions (meaning adverse effects on human health, such as death, disability, illness or disorders) caused by pathogenic micro-organisms, protozoa and bacteria, intestinal parasites, viruses that are transmitted in water. These diseases can be spread while bathing, washing, drinking water, or by eating food exposed to contaminated water. While diarrhea and vomiting are the most commonly reported symptoms of waterborne illness, other symptoms can include skin, ear, respiratory, or eye problems. Lack of clean water supply, sanitation and hygiene (WASH) are major causes for the spread of waterborne diseases in a community. Therefore, reliable access to clean drinking water and sanitation is the main method to prevent waterborne diseases.	**7**	**CO .2**	**K 3**

Disease and transmission[3][9]	Microbial agent	Sources of agent in water supply	General symptoms
Giardiasis (fecal-oral) (hand-to-mouth)	Protozoan (Giardia lamblia) Most common intestinal parasite	Untreated water, poor disinfection, pipe breaks, leaks	Diarrhea, ab discomfort, bloating and flatulence
Cholera	Spread by the bacterium Vibrio cholerae	Drinking water contaminated with the bacterium	rapidly fatal known, diarrhea, nausea , nosebleed, rap vomiting, and hypovolemi shock (in severe at which point d occur in 12–18 h
E. coli Infection	Certain strains of Escherichia coli (commonly *E . coli*)	Water contaminated with the bacteria	Mostly diarrhe cause in immunocomp individuals, th young, and the due to dehydrat prolonged illnes
Typhoid fever	Salmonella typhi	Ingestion of water contaminated with feces of an infected person	fever up to (104 °F), sweati diarrhea, muscl fatigue, con spleen, liver enl t, cause deat called "rose small red spots abdomen and ch
Hepatitis A	Hepatitis A virus (HAV)	Can manifest itself in water (and food)	Symptoms only acute (no chronic sta virus) and include fever, malaise, abdomin nausea, diarrhea, weig

			itching, jaundice, and depression.
Hepatitis E (fecal-oral)	Hepatitis E virus (HEV)	Enters water through the feces of infected individuals	acute hepatitis (liver disease), including fever, fatigue, loss of appetite, nausea, vomiting, abdominal pain, jaundice, dark urine, clay-colored stool, and joint pain
Poliomyelitis (Polio)	Poliovirus	Enters water through the feces of infected individuals	headache, fever, and occasional seizures, and spastic paralysis, 1% have symptoms of non-paralytic aseptic meningitis. The rest have serious symptoms resulting in paralysis or death

Rural populations are more at risk from waterborne illnesses, but everyone faces risks of polluted or contaminated water. Waterborne illness can affect anyone, anywhere. The risk is more for infants, younger children, the elderly and patients of diabetes, chronic diseases of heart disease, kidney, etc.

Precautions to prevent waterborne disease:
Ensure the water is visibly clean and free from sand and silt. Filter the water to get rid of visible dirt.

Get water purifying devices like filters, RO unit, etc., regularly serviced and maintained.

Hand hygiene – regularly wash hands with soap after returning home, after using the toilet, before and after preparing food, before eating or drinking anything.

Teach hand hygiene to children. Children should make it a habit to always wash hands when returning home after playing games.

Take vaccinations for immunization against preventable diseases like Typhoid, Hepatitis A, Polio, etc.

Water induced diseases
Vector Borne diseases are the illness caused by the vectors. A vector is a carrier of the causative microbe for various diseases such as mosquitoes, ticks and fleas

Example; Malaria, dengue, Chikingunia, Japanese Encaphilitis, west Nile fever, yellow fever

Malaria
Pathogen- malaria is caused by ***Plasmodium a tiny protozoan***, which are spread to people through the bites of infected female *Anopheles* mosquitoes.

Causes

Malaria is caused by a single-celled parasite of the genus plasmodium. The parasite is transmitted to humans most commonly through mosquito bites.

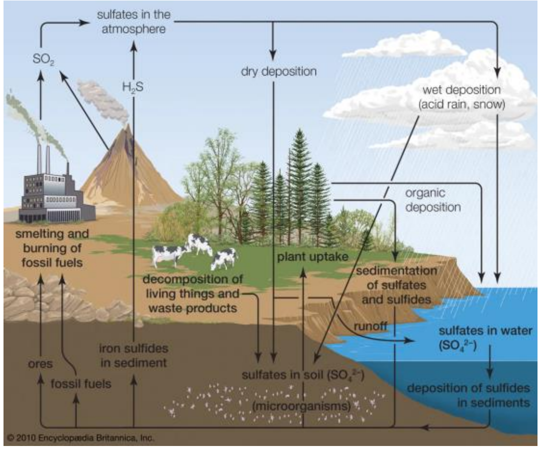
Symptoms- Signs and symptoms of malaria may include: High Fever, Chills, Headache, Nausea and vomiting, Abdominal pain, Muscle or joint pain, Convulsion,

	Prevention Mosquitoes are most active between dusk and dawn. To protect yourself from mosquito bites, you should: Cover your skin, **Apply insect repellent to skin, Apply repellent to clothing, Sleep under a net.** ***<u>Dengue/Backbone fever</u>*** Dengue virus is transmitted by female mosquitoes mainly of the species ***Aedes aegypti*****.** *Aedes. aegypti* is a day-time feeder; its peak biting periods are early in the morning Symptoms 1) Bleeding gums, Persistent vomiting, Rapid breathing, Abdominal pain, Blood in vomit **Prevention** • Disposing of solid waste properly and removing artificial man-made habitats that can hold water. • Using of personal household protection measures, such as window screens, repellents, coils and vaporizers. • Wearing clothing that minimizes skin exposure to mosquitoes is advised. ***<u>Yellow fever</u>*** Yellow fever is an acute viral hemorrhage disease transmitted by ***<u>aedes aegypti</u>*** mosquitoes. **Symptoms,** High Fever, Headache, Jaundice, muscle pain, Vomiting, Backpain, Skin bleeding ***<u>Chikingunia</u>*** Chikungunya virus is transmitted between humans via mosquitoes. When a naïve (uninfected) mosquito feeds upon a viremic person (someone who has the virus circulating in their blood), the mosquito can pick up the virus as it ingests the blood. **Symptoms** Muscle pain, Joint swelling, Joint Pain, Fever, Headache, Nausea, Fatigue and rash. **Prevention** Do not store water in open containers so that they do not become breeding sites for mosquitoes, Cover tanks or containers for water for domestic use, Use mosquito repellent or long sleeves to avoid getting bitten. **<u>Zika virus</u>** Zika virus disease is caused by a virus transmitted primarily by *Aedes mosquitoes*, which bite during the day. It also results in fetal loss, preterm birth and miscarriage. An increased risk of neurologic complications is associated with Zika virus infection in adults and children, including Guillain-Barré syndrome and neuropathy **Symptoms,** Mild and include fever, rash, conjunctivitis, muscle and joint pain, malaise or headache. **Prevention** Applying insect repellent to skin, Removing standing water from flower pots, Cleaning up trash and used tyres, Use of insecticides to reduce mosquito population, Young children and pregnant women should sleep under mosquito nets.			
Q 2	Write short notes on (a) Overnutrition . Overnutrition is a type of malnutrition that occurs when the body consumes more nutrients than it needs. It can lead to weight gain, obesity, and other health problems. Health effects	**7**	**CO.2**	**K3**

	- Non-communicable diseases (NCDs) like cardiovascular disease, type 2 diabetes, and stroke, Chronic inflammation, Metabolic disorders, Dyslipidemia, Hypertension Environmental impact - The production and consumption of excess food puts pressure on natural resources and the environment (b)Global Water distribution The distribution of water on Earth is uneven, with most of the water being saline and stored in the oceans. The remaining water is found in glaciers, polar ice caps, rivers, lakes, and groundwater. Water distribution **Oceans**: Contain about 97% of the Earth's water, which is too salty for drinking, growing crops, and most industrial uses **Glaciers and ice caps**: Contain about two-thirds of the Earth's freshwater **Groundwater**: Contains about a third of the Earth's freshwater **Rivers and lakes**: Contain less than 1% of the Earth's freshwater			
Q 3	Discuss about various problems due to biological and chemical contamination of water. Biological contamination of water is responsible for Water Borne and water induced diseases. Chemical contamination of water: Fluorosis is a condition that occurs when someone consumes too much fluoride over a long period of time. It can affect the teeth and bones. Dental fluorosis - A change in the appearance of teeth caused by too much fluoride during tooth development - Appears as white flecks, spots, or lines on the enamel - Can also appear as yellow, brown, or black discoloration - Can make teeth rough and hard to clean - Only occurs in children because permanent teeth are fully formed by age 8 - In most cases, it's so mild that it's only noticeable by a dentist - Doesn't affect tooth health or function Skeletal fluorosis - A condition that causes extra bone growth, which can lead to bone density increases and joint enlargement - Can cause bone fractures, spinal deformities, and other symptoms Treatments - Mild cases may not require treatment - Moderate to severe cases can be treated with tooth whitening, bonding, crowns, veneers, or MI Paste Fluorosis can be caused by drinking water with high levels of fluoride, or by	**7**	**CO .2**	**K 3**

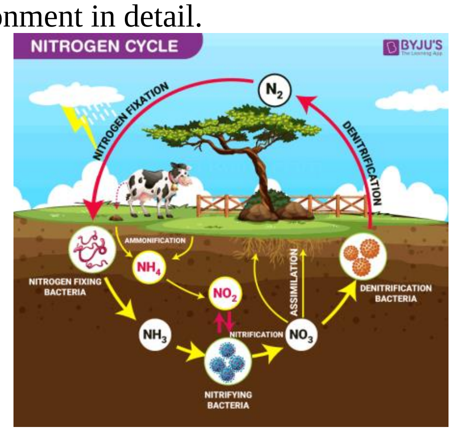
	swallowing too much fluoride toothpaste.			
Q 4	Write short note on Water logging **Water logging** Waterlogging is the condition of having too much water in the soil, especially in the root zone of plants. It can be caused by a number of factors, including high groundwater tables, heavy rainfall, and poor drainage. **Causes** **Groundwater table:** When the groundwater table is too high, it can prevent air from reaching the soil. **Rainfall:** Heavy rainfall can cause waterlogging, especially in areas with poor drainage. **Irrigation:** Irrigation water can introduce salts into the soil, which can lead to soil salinity. **Soil type:** Different types of soil have different water permeability, which can affect how much water they absorb. **Effects** **Oxygen deficiency**: Waterlogging can reduce the amount of oxygen available to plant roots, which can lead to plant death. **Nutrient deficiencies**: Waterlogging can cause nutrient deficiencies or toxicities in plants. **Soil salinity**: Waterlogging can lead to soil salinity, which can be a long-term water quality concern. **Solutions** **Land management:** Land management techniques like contour farming, terracing, and raised beds can help reduce waterlogging. **Crop rotation:** Different crops have different water requirements, so rotating crops can help prevent waterlogging. **Crop selection** Choosing crops that are more tolerant to waterlogging can help prevent waterlogging (ii) Shifting cultivation **Shifting cultivation** is a traditional agricultural technique that involves clearing land, burning vegetation, and growing crops for a short time before moving on to a new plot. It's also known as slash-and-burn or swidden agriculture. Steps 1. Select a plot of land 2. Cut down trees and bushes 3. Burn the vegetation to clear the land 4. Grow crops for a short time 5. Move on to a new plot of land and repeat the process Benefits • Allows the land to regenerate and maintain its fertility Drawbacks • Can lead to deforestation and soil degradation • Can cause uncontrollable forest fires	**7**	**CO .2**	**K 3**

	• Releases a significant amount of carbon emissions • Robs the soil of its nutrients, making it infertile			
Q 5	Describe various kinds of energy resources. Distinguish between conventional and non-conventional energy resources. There are many types of energy resources, including renewable and non-renewable resources. Renewable energy resources/Non-conventional • **Solar energy**: Sunlight is converted into electricity using photovoltaic cells. • **Wind energy**: Wind turbines harness the kinetic energy of moving air to generate electricity. • **Hydropower**: The movement of water, usually by dams or water turbines, generates electricity. • **Geothermal energy**: Heat from the Earth's interior is used to generate electricity and heat. • **Biomass energy**: Organic materials like wood, agricultural waste, and algae are burned or converted into biofuels to produce heat and electricity. • **Tidal energy**: The up and down motion of floating devices on the ocean's surface converts the energy from tides into electricity. Non-renewable energy resources/Conventional • **Fossil fuels**: Oil, coal, and natural gas are examples of fossil fuels.	**7**	**CO .2**	**K 3**
Q 6	Draw and explain the circulation of Sulphur in the environment. The sulphur cycle is the process by which sulphur moves through the atmosphere, living things, and mineral forms. sulphur is a vital component of many proteins and is found in seawater and sedimentary rocks. Steps in the sulphur cycle • **Decomposition of organic compounds**: The breakdown of proteins releases amino acids that contain sulfur. • **Oxidation of hydrogen sulfide to elemental** sulphur: Photosynthetic bacteria like Chlorobiaceae and Chromatiaceae oxidize hydrogen sulfide to produce elemental sulphur • **Oxidation of elemental** sulphur: Chemolithotrophic bacteria convert elemental sulphur into sulphates because plants can't use it directly. • **Reduction of sulphates**: Desulfovibrio desulfuricans reduces sulphates to hydrogen sulfide. • **Sulfide oxidation**: Sulfide oxidation occurs throughout the sediment column, but is most active in the surface zone.	**7**	**CO .2**	**K 3**



Q 7	How does deforestation cause deterioration of the quality of life in a tribal society. Key impacts on tribal populations due to deforestation: **Loss of habitat and food sources:** Forests provide tribal communities with food, medicine, and other essential resources, which are drastically reduced when trees are cleared, leaving them with limited options for sustenance. **Displacement and migration:** As their land is taken over for development, tribal people are often forced to relocate, disrupting their social structures and cultural practices. **Erosion of cultural identity:** Forests are often deeply intertwined with tribal beliefs and traditions, and their loss can lead to a decline in cultural knowledge and practices. **Increased vulnerability to climate change:** Deforestation disrupts the local climate, leading to more extreme weather events which disproportionately affect tribal communities with limited resources to adapt. **Conflict and exploitation:** When forests are cleared for commercial activities, tribal communities can face conflicts with outside interests, leading to exploitation and human rights abuses. **Health issues:** Reduced access to traditional medicines and clean water due to deforestation can negatively impact the health of tribal populations.	**7**	**CO .2**	**K 3**
Q 8	What do you mean by forest resources? Explain the direct and indirect benefits of forest resources. Explain the forest conservation and management in detail. Forest resources offer several advantages like providing oxygen, regulating climate, protecting watersheds, supporting biodiversity, and offering recreational opportunities, but also have disadvantages including potential for deforestation, soil erosion, habitat loss for wildlife, and disruption of ecosystems when harvested unsustainably. Advantages of forest resources: **Environmental benefits:** **Oxygen production:** Trees absorb carbon dioxide and release oxygen, contributing to atmospheric balance. **Climate regulation:** Forests help regulate temperature and humidity by absorbing heat and releasing moisture. **Water conservation:** Tree canopies intercept rainfall, preventing rapid runoff and helping to replenish groundwater. **Soil erosion control:** Tree roots hold soil in place, preventing erosion. **Biodiversity conservation:** Forests provide habitats for a wide variety of plant and animal species. **Economic benefits:**	**7**	**CO .2**	**K 3**

	Timber production: Forests provide wood for construction, furniture, and other products. **Non-timber products:** Forests yield fruits, nuts, medicinal plants, resins, and other valuable resources. **Ecotourism:** Forests can attract tourists for activities like hiking, camping, and wildlife viewing. **Social benefits:** **Recreation:** Forests provide spaces for outdoor activities like hiking, camping, and fishing. **Cultural significance:** Many cultures have strong spiritual and cultural ties to forests. ***Forest conservation and management*** is the practice of protecting and restoring forests to ensure their long-term health and sustainability. The goal is to maintain the balance of species and gene pools, and to reduce the effects of deforestation. Methods **Sustainable land management**: Uses forest resources in a way that is environmentally responsible, economically viable, and socially acceptable **Reforestation**: Planting trees to replace those that have been lost **Agroforestry**: Integrating trees and shrubs into agricultural landscapes **Community-based forest management**: Involving local communities in the management of forest resources **Forest fire prevention**: Taking measures to prevent and control forest fires **Regulating grazing**: Limiting the number of animals that graze on forest land **Protecting from pests and pathogens**: Taking steps to protect forests from pests and diseases **Educating the public**: Spreading awareness about the importance of forests and the consequences of deforestation			
Q 9	With the help of well labeled sketch of nitrogen cycle, explain how elemental nitrogen is recycled in nature. Nitrogen cycle is an important part of the ecosystem. In this article, we shall explore its implications on the environment in detail.	**7**	**CO .2**	**K 3**



"Nitrogen Cycle is a biogeochemical process which transforms the inert nitrogen present in the atmosphere to a more usable form for living organisms."

Process of the Nitrogen Cycle consists of the following steps – Nitrogen fixation, Nitrification, Assimilation, Ammonification and Denitrification. These processes take place in several stages and are explained below:

Nitrogen Fixation Process

Here, Atmospheric nitrogen (N_2) which is primarily available in an inert form, is converted into the usable form -ammonia (NH_3). During the process of Nitrogen fixation, the inert form of nitrogen gas is deposited into soils from the atmosphere and surface waters, mainly through precipitation. The entire process of Nitrogen fixation is completed by symbiotic bacteria, which are known as Diazotrophs. *Azotobacter* and *Rhizobium* also have a major role in this process.

Nitrification

In this process, the ammonia is converted into nitrate by the presence of bacteria in the soil. Nitrites are formed by the oxidation of ammonia with the help of *Nitrosomonas* bacteria species. Later, the produced nitrites are converted into nitrates by *Nitrobacter*. This conversion is very important as ammonia gas is toxic for plants.

Assimilation

Primary producers – plants take in the nitrogen compounds from the soil with the help of their roots, which are available in the form of ammonia, nitrite ions, nitrate ions or ammonium ions and are used in the formation of the plant and animal proteins.

Ammonification

When plants or animals die, the nitrogen present in the organic matter is released back into the soil. The decomposers, namely bacteria or fungi present in the soil, convert the organic matter back into ammonium. This process of decomposition produces ammonia, which is further used for other biological processes.

Denitrification

Denitrification is the process in which the nitrogen compounds make their way back into the atmosphere by converting nitrate (NO_3-) into gaseous nitrogen (N). This process of the nitrogen cycle is the final stage and occurs in the absence of oxygen. Denitrification is carried out by the denitrifying bacterial species- *Clostridium* and *Pseudomonas*, which will process nitrate to gain oxygen and gives out free nitrogen gas as a byproduct.

Q 10 Differentiate conventional and non-conventional energy resources. Explain the working principle, advantages, disadvantages of nuclear energy. Also explain various types of nuclear reactors with neat sketch. **7** **CO.2** **K3**

Difference Between Conventional and Non-conventional Sources of Energy

Conventional Sources of Energy	Non-conventional sources of e
These sources of energy are also known as a non-renewable source of energy	These sources of energy are al renewable source of energy
They find both commercial and industrial purposes	They are mainly used for housel
These can be considered to be one of the reasons for the cause of pollution	These are not responsible fo pollution

Coal, fossil fuels are two examples	Wind, solar energy and Biomass two examples

Nuclear energy works by splitting atoms in a nuclear reactor to create heat, which is then used to produce electricity. The process of splitting atoms is called fission, and the heat released from fission is used to boil water and create steam. The steam then spins a turbine, which activates a generator to produce electricity.

How it works

Fission: A neutron collides with a large atom, causing it to split into two smaller atoms. This process releases a large amount of energy.

Chain reaction: The splitting of one atom releases additional neutrons that can split other atoms, creating a chain reaction.

Heat: The heat from fission is used to boil water and create steam.

Turbine: The steam spins a turbine.

Generator: The spinning turbine activates a generator, which produces electricity.

Nuclear energy has many advantages, including:

Low carbon emissions: Nuclear power is a clean energy source that produces no greenhouse gases.

Reliable: Nuclear power plants can operate for years without interruption.

Affordable: Nuclear power is cheaper than coal and natural gas.

Creates jobs: The nuclear industry employs many people.

Space exploration: Nuclear power is used in radioisotope power systems (RPSs) to explore deep space.

Medical diagnosis and treatment: Nuclear power is used in medical diagnosis and treatment.

Criminal investigation: Nuclear power is used in criminal investigation.

Agriculture: Nuclear power is used in agriculture.

Small land footprint: Nuclear power produces more electricity on less land than other clean-air sources.

Dense fuel: Nuclear fuel is extremely dense.

There are several types of nuclear reactors, including pressurized water reactors (PWRs), boiling water reactors (BWRs), and light water graphite-moderated reactors (LWGRs). Other types include gas-cooled reactors, fast reactors, and molten salt reactors.

Pressurized water reactors (PWRs)

- The most common type of nuclear reactor in the world
- Uses water as a coolant and moderator
- Keeps water under high pressure to prevent it from boiling
- Transfers heat to a secondary loop that produces steam to drive turbines

Boiling water reactors (BWRs)

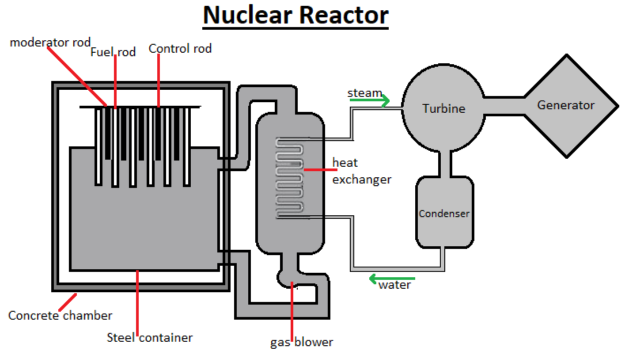
- Allows water to boil, carrying away heat generated in the fuel rods as steam

Light water graphite-moderated reactors (LWGRs) Uses graphite as a moderator.

Other types of nuclear reactors

- **Pressurized heavy water reactors (PHWRs)**: Uses heavy water as a moderator
- **Fast reactors**: Developed to a full-scale demonstration stage
- **Molten salt reactors**: A type of nuclear reactor
- **Small modular reactors (SMRs)**: Smaller units that can be built independently or as modules in a larger complex
- **Nuclear fusion reactors**: A low-carbon energy source that uses the process of

fusion to generate heat and light



Q 11 What do you mean by defluoridation? Discuss about Nalgonda technique in this context. **7** **CO.2** **K3**

. "Defluoridation" refers to the process of removing excess fluoride from water, while the "Nalgonda technique" is a specific method used for defluoridation, which involves adding chemicals like aluminum salts, lime, and bleaching powder to water to precipitate and remove fluoride through flocculation and sedimentation; essentially, it's a widely used, cost-effective technique for defluoridation, particularly in areas with high fluoride levels in India.

The Nalgonda technique is a method for treating water to remove fluoride, color, odor, turbidity, bacteria, and organic contaminants. It's often used in rural areas to treat groundwater that has high levels of fluoride.

How it works

1. Aluminium salts (aluminium sulfate-alum or aluminium chloride or combination of these two) : responsible for removal of fluoride from water
2. Lime : facilitates forming dense flocks for rapid settling of insoluble fluoride salts .
   Dose : empirically 1/20th of that of the dose of aluminium salt
3. Bleaching powder : disinfection-3 mg/l

Mechanism of Nalgonda technique

–Rapid mix:- provides thorough mixing of chemicals
–Flocculation:- gentle agitation
- combination of poly hydroxy aluminum complex with fluoride & polymeric aluminum hydroxides are formed (flocs)
- turbidity, color ,odour → removed; bacterial load reduces
- Lime ensures that residual aluminum does not remain in treated water

- Sedimentation:- permits settling of flocs loaded with fluorides & other impurities
- Filtration:-rapid gravity sand filters
- Disinfection :- rechlorinated with bleaching powder before distribution

Advantages of Nalgonda technique

- Regeneration of media is not required.
- No handling of caustic acids and alkalies.
- The chemicals required are readily available and are used in conventional municipal water treatment.
- Adaptable to domestic use.
- Economical
- Can be used to treat water in large quantities for community usage.

- Applicable in batch as well as in continuous operation to suit needs.
- Simplicity of design, construction, operation and maintenance.
- Local semi-skilled workers can be readily employed.
- Highly efficient removal of fluorides from high levels to desirable levels
- Simultaneous removal of color, odor, turbidity, bacteria and organic contaminants.

- Little wastage of water and least disposal problems.
- Needs minimum of mechanical and electrical equipment.
- No energy, except muscle power is required for domestic equipment.
- Provides de-fluoridated water of uniform acceptable quality.

Disadvantages of Nalgonda technique

- Desalination may be necessary when the total dissolved solids exceed 1500 mg/l.
- Generation of higher quantity of sludge
- The large amount of alum needed to remove fluoride.
- Careful pH control of treated water is required.
- High residual aluminium is reported in treated water by some authors.

Q 12 What are the effects of mining on the well-being of minors and environment? **7** **CO.2** **K3**

Mining can negatively impact the environment and the health of miners.

Environmental effects

- **Water pollution**: Mining can pollute water with heavy metals, like arsenic, copper, lead, and zinc, and other chemicals. This can harm aquatic life.
- **Soil erosion**: Mining can cause soil erosion, which can lead to sinkholes and mudslides.
- **Loss of biodiversity**: Mining can lead to the loss of biodiversity, especially if forests are removed and not restored.
- **Air pollution**: Mining can emit carbon into the atmosphere, which contributes to climate change.
- **Acid drainage**: When sulfide minerals like pyrite are exposed to air and water, they react to create sulfuric acid, which pollutes water.
- **Noise pollution**: Mining can create noise pollution.

Health effects

- **Respiratory issues**: Mining can cause respiratory issues.
- **Physical injuries**: Mining can cause physical injuries.
- **Exposure to hazardous substances**: Mining can expose miners to hazardous substances.

Q 13 Name, explain and draw the various steps carried out in carbon cycle. **7** **CO.2** **K3**

Carbon Cycle Definition

Carbon cycle is the process where carbon compounds are interchanged among the biosphere, geosphere, pedosphere, hydrosphere, and atmosphere of the earth.

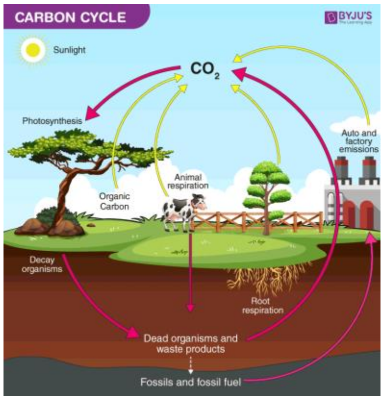
Carbon Cycle Steps

Following are the major steps involved in the process of the carbon cycle:

1. Carbon present in the atmosphere is absorbed by plants for photosynthesis.
2. These plants are then consumed by animals and carbon gets bioaccumulated into their bodies.
3. These animals and plants eventually die, and upon decomposing, carbon is released back into the atmosphere.
4. Some of the carbon that is not released back into the atmosphere eventually become fossil fuels.
5. These fossil fuels are then used for man-made activities, which pump more carbon back into the atmosphere.

Carbon Cycle Diagram

The carbon cycle diagram below elaborates the flow of carbon along different paths.



Carbon Cycle diagram showing the flow of carbon, its sources and paths.

Carbon Cycle on Land

Carbon in the atmosphere is present in the form of carbon dioxide. Carbon enters the atmosphere through natural processes such as respiration and industrial applications such as burning fossil fuels. The process of photosynthesis involves the absorption of CO_2 by plants to produce carbohydrates. The equation is as follows:

$CO_2 + H_2O + \text{energy} \rightarrow (CH_2O)n + O_2$

Carbon compounds are passed along the food chain from the producers to consumers. The majority of the carbon exists in the body in the form of carbon dioxide through respiration. The role of decomposers is to eat the dead organism and return the carbon from their body back into the atmosphere. The equation for this process is:

$(CH_2O)n + O_2 \rightarrow CO_2 + H_2O$

Importance of Carbon Cycle

Even though carbon dioxide is found in small traces in the atmosphere, it plays a vital role in balancing the energy and traps the long-wave radiations from the sun. Therefore, it acts like a blanket over the planet. If the carbon cycle is disturbed it will result in serious consequences such as climatic changes and global warming.

Carbon is an integral component of every life form on earth. From proteins and lipids to even our DNA. Furthermore, all known life on earth is based on carbon. Hence, the carbon cycle, along with the nitrogen cycle and oxygen cycle, plays a vital role in the existence of life on earth.

Key Points on Carbon Cycle

- Carbon cycle explains the movement of carbon between the earth's biosphere, geosphere, hydrosphere and atmosphere.
- Carbon is an important element of life.
- Carbon dioxide in the atmosphere is taken up by green plants and other photosynthetic organisms and is converted into organic molecules that travel through the food chain. Carbon atoms are then released as carbon dioxide when organisms respire.
- The formation of fossil fuels and sedimentary rocks contributes to the carbon cycle for very long periods.
- The carbon cycle is associated with the availability of other compounds as well.

Q 14	Enumerate the application of solar energy in modern days. Solar energy has many applications, including heating, power generation, and cooking. Heating	**7**	**CO .2**	**K 3**

	• **Solar water heaters**: Solar energy can be used to heat water for homes and other buildings **Solar space heating**: Solar thermal energy can be used to heat spaces like greenhouses, swimming pools, and livestock buildings **Solar cooking**: Solar energy can be used for cooking Power generation • **Solar power plants**: Solar energy can be used to generate electricity for homes and businesses • **Solar ponds**: Solar ponds can be used to store solar energy as heat, which can then be used to generate electricity • **Photovoltaic panels**: Solar panels can be used to generate electricity for homes and businesses Other applications • **Charging batteries**: Solar electricity can be used to charge batteries • **Solar-powered pumps**: Solar energy can be used to power pumps • **Solar cars**: Solar energy can be used to power vehicles • **Desalination**: Solar energy can be used to desalinate water • **Industrial process heating**: Solar energy can be used to heat fluids for industrial processes			
Q 15	Explain the fluoride problem in drinking water. What measures should be taken to control the fluoridation? ***<u>Fluorides in drinking water</u>*** Fluoride is the one of the few chemicals that has been shown to cause significant effects in people though drinking water. Fluoride has benefical effects on teeth at low concentration in drinking water but excessive exposure to fluorides in drinking water can give rise to number of adverse effects. In 20 states of India, more than 100 districts across the country and probably more than 60 million people are consuming drinking water which has fluoride greater than 1 mg/l. Since local food can also get irrigated by the same water, food also contains fluoride in these places. This makes the total daily consumption of fluoride more than 10 mg/day which is always harmful for adults and more so for children. Firstly, the body requirement for calcium increases. This makes specific people such as women in pregnancy and lactation, growing children and adults beyond the age of 40, more prone to calcium related problems. Apart from this, iron absorption is reduced due the fluoride. This is really important to all of us and especially for pregnant women for whom iron deficiency anemia and related problems are a serious cause of under-weight and unhealthy children at birth. In ground water, however low or high concentration of fluorides can occur depending upon the nature of rocks and the occurrence of fluoride bearing minerals. *<u>Fluoride (range)</u>* • Low level of fluoride in drinking water 0.5 to 1mg/l protect against dental caries • High level of fluoride in drinking water (above 1.5mg/l) will lead to adverse health impacts ranging from dental fluorosis to skeletal fluorosis • Fluoride levels in water beyond desirable /permissible limits is typically found in ground water and not surface water. *<u>Effects on human</u>* The effects of long term exposure to naturally occurring fluoride from drinking water and other environmental sources are the major concern with regards to human health. *<u>Effects on teeth</u>*	**7**	**CO .2**	**K 3**

	High level of fluoride upto $10mgl^{-1}$ were associated with dental fluorosis. (yellowish or brownish striations or mottling of enamel) while low level of fluoride less than $0.1mgl^{-1}$ were associated with high level of dental decay, although poor nutritional status is also an important contribution factor. *Skeletal effects* Endemic skeletal fluorosis is primarily associated with the consumption of drinking water containing elevated level of fluoride but exposure to additional sources of fluoride such as high fluoride coal is also potentially very important. This is compounded by a number of factors which include climate, related to water consumption, nutritional status and diet. Crippling skeletal fluorosis which is associated with the higher level of exposure, can result from osteosclerosis, ligamentous and tendinous calcification and extreme bone deformity. *Symptoms* Fatique, Low HB count, Irritation, Loss of appetite, Excess thirst and urination, Miscarriage Inability to conceive due to detective sperm, Depression *Impact of skeletal fluorosis* Loss of livelihood since affected persons cannot work, Loss of income, High medical costs Poverty, Social outcosts since facilities for the handicapped are lacking in india *Suggestions* Eat more of fruits and vegetables rich in vitamin C, iron and calcium			

Unit 3

UNIT 3
SOLUTIONS
1. Differentiate BOD & COD?

- **BOD (Biochemical Oxygen Demand):** Measures oxygen consumed by microorganisms to decompose organic matter in water over 5 days (BOD_5). Reflects biodegradable organic content.
- **COD (Chemical Oxygen Demand):** Measures total oxygen required to chemically oxidize organic and inorganic matter using strong oxidizing agents (e.g., potassium dichromate). Provides faster results (hours) and includes non-biodegradable substances.

2. What is sludge treatment?
Sludge treatment processes reduce volume, stabilize organic matter, and prepare waste for disposal/reuse. Steps include:

- **Thickening:** Concentrates sludge.
- **Digestion:** Anaerobic/aerobic breakdown to reduce pathogens and odor.
- **Dewatering:** Removes water (e.g., centrifugation, filtration).
- **Disposal/Reuse:** Incineration, land application, or composting.

3. Describe the **Types of Carbon Cycle**

- **Biological Cycle:** Short-term; involves photosynthesis, respiration, and decomposition.
- **Geological Cycle:** Long-term; includes rock weathering, sedimentation, and volcanic CO_2 release.
- **Oceanic Cycle:** CO_2 exchange between atmosphere and oceans; marine organisms use carbon for shells.
- **Anthropogenic Influence:** Fossil fuel combustion and deforestation disrupt natural cycles.

4. Define Photochemical Smog
Air pollution formed when sunlight reacts with nitrogen oxides (NO_x) and volatile organic compounds (VOCs). Produces ozone (O_3), peroxyacyl nitrates (PANs), and haze. Common in urban areas with high vehicle emissions.

5. Define the **Carbon Credits**
Permits allowing emission of 1 ton of CO_2-equivalent. Used in cap-and-trade systems to incentivize emission reductions. Companies can trade credits or invest in offsets (e.g., reforestation) to meet regulatory targets.

6. Define Effluent
Treated or untreated wastewater discharged from industrial, agricultural, or domestic sources into water bodies. Regulated to minimize environmental harm.

7. How is domestic discharge responsible for increasing the BOD level into the stream?
Domestic wastewater contains organic waste (food, feces, detergents). When released untreated, microbial decomposition consumes dissolved oxygen, raising BOD. This depletes oxygen, harming aquatic life (eutrophication).

8. Define Freshwater
Water with low dissolved salts (<1% salinity). Sources include rivers, lakes, groundwater, and glaciers. Critical for drinking, agriculture, and ecosystems. Contrasts with saline (ocean) and brackish water.

Long Questions
1. Typical Layout of a Conventional Sewage Treatment Plant:
A conventional sewage treatment plant is designed to remove physical, chemical, and biological contaminants from wastewater. The layout typically includes the following units:
Preliminary Treatment

- **Screening**
  - **Function:** Removes large debris (e.g., sticks, rags, plastics) using bar screens or mechanical screens.
  - **Purpose:** Protects downstream equipment from clogging or damage.
- **Grit Chamber**
  - **Function:** Settles heavy inorganic particles like sand, gravel, and grit through reduced flow velocity.
  - **Purpose:** Prevents abrasion in pumps and pipes.

2. Primary Treatment

- **Primary Sedimentation Tank (Primary Clarifier)**
  - **Function:** Allows suspended solids to settle by gravity, forming **primary sludge**.
  - **Purpose:** Removes ~50-60% of suspended solids and ~30% of organic matter (BOD).

    - **Design:** Circular or rectangular tanks with slow-flowing wastewater.

3. Secondary Treatment (Biological Treatment)

- **Aeration Tank (Activated Sludge Process)**
    - **Function:** Aerobic microorganisms break down dissolved organic matter using oxygen (supplied via blowers or mechanical aerators).
    - **Purpose:** Reduces BOD and converts organic matter into CO_2, water, and microbial biomass.
- **Secondary Sedimentation Tank (Secondary Clarifier)**
    - **Function:** Settles the biological sludge (activated sludge) formed in the aeration tank.
    - **Purpose:** Separates treated water from microbial biomass. Part of the sludge is recycled to the aeration tank.

4. Tertiary Treatment (Optional, Advanced Treatment)

- **Filtration**
    - **Function:** Removes fine suspended particles using sand filters, membrane filters, or disc filters.
- **Nutrient Removal**
    - **Process:** Removes nitrogen (via nitrification-denitrification) and phosphorus (via chemical precipitation).
- **Disinfection**
    - **Methods:** Chlorination, UV radiation, or ozonation.
    - **Purpose:** Kills pathogens (bacteria, viruses) before discharge.

5. Sludge Treatment

- **Sludge Digestion**
    - **Anaerobic Digestion:** Breaks down organic sludge in oxygen-free tanks, producing biogas (methane).
    - **Aerobic Digestion:** Uses oxygen to stabilize sludge.
- **Sludge Dewatering**
    - **Methods:** Centrifuges, filter presses, or drying beds reduce water content.
- **Disposal/Reuse**
    - **Options:** Landfill, incineration, or agricultural use as fertilizer (if treated).

Flow Diagram of the Process

1. **Influent** → Screening → Grit Chamber → Primary Sedimentation → Aeration Tank → Secondary Sedimentation → Tertiary Treatment → Disinfection → **Effluent** (to water bodies).

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2. Causes and Effects of Soil Erosion:Causes:

- Deforestation: Removal of trees reduces root systems that hold soil together.
- Overgrazing: Excessive grazing by livestock removes vegetation cover.
- Improper Farming Practices: Plowing, monoculture, and lack of crop rotation.
- Construction Activities: Disturbs soil structure and removes vegetation.
- Natural Events: Heavy rainfall, floods, and wind.

Deforestation- Trees and vegetation act as natural anchors for soil. Their roots bind soil particles together, creating a stable structure.

- **Impact of Removal:**
    - Loss of root systems weakens soil cohesion, making it vulnerable to being washed or blown away.
    - Reduced canopy cover exposes soil to direct rainfall and wind.
- **Example:** In the Amazon rainforest, deforestation for agriculture or logging has led to severe soil erosion, reducing fertility and increasing sedimentation in rivers.

Overgrazing- Livestock (e.g., cattle, sheep) consume vegetation faster than it can regrow, stripping the land of protective plant cover.

Impact: Bare soil is exposed to wind and water erosion, Animal hooves compact soil, reducing its ability to absorb water and increasing surface runoff, **Example:** Overgrazing in arid regions like the Sahel (Africa) has turned once-productive land into deserts (desertification).

Improper Farming Practices

Plowing:

- Tilling soil disrupts its structure, breaking down aggregates and making it prone to erosion.
- Plowing along slopes creates channels for water to carry soil away (rill and gully erosion).

Monoculture:

- Repeatedly growing the same crop depletes specific nutrients, weakening soil health.
- Lack of diverse root systems reduces soil stability.

Lack of Crop Rotation:

- Prevents soil from recovering organic matter, leading to nutrient depletion and reduced resistance to erosion.

Example: The 1930s Dust Bowl in the U.S. was worsened by intensive plowing of prairie grasslands for wheat farming.

4. Construction Activities- Clearing vegetation for buildings, roads, or mining removes protective plant cover.

- Excavation and earth-moving disrupt soil layers, loosening particles.

- **Impact:**
  - Exposed soil is easily washed away by rain or carried by wind.
  - Compacted soil from machinery reduces infiltration, increasing runoff.
- **Example:** Urban sprawl in hilly areas often leads to landslides and sediment pollution in nearby water bodies.
- **Effects:**
  - Loss of fertile topsoil, reducing agricultural productivity.
  - Sedimentation in rivers and lakes, affecting aquatic ecosystems.
  - Increased flooding due to reduced water infiltration.
  - Habitat destruction and loss of biodiversity.

1) **Loss of Fertile Topsoil, Reducing Agricultural Productivity**

Topsoil is the uppermost layer of soil, rich in organic matter, nutrients (nitrogen, phosphorus, potassium), and microorganisms essential for plant growth. Erosion strips away this layer.

- **Impact:**
  - **Reduced Crop Yields:** Loss of nutrients forces farmers to rely on synthetic fertilizers, increasing costs and pollution.
  - **Soil Degradation:** Erosion leaves behind compacted, nutrient-poor subsoil, which is less fertile and harder to cultivate.
  - **Economic Loss:** Globally, soil erosion costs agriculture **$400 billion annually** in lost productivity (UN Food and Agriculture Organization).
- **Example:** In India's Punjab region, intensive farming and erosion have degraded 75% of the soil, threatening food security.

2. Sedimentation in Rivers and Lakes, Affecting Aquatic Ecosystem- Eroded soil particles (sediment) are carried by runoff into water bodies.

- **Impact:**
  - **Clogged Gills:** Suspended sediment suffocates fish and aquatic invertebrates by blocking their respiratory systems.
  - **Reduced Light Penetration:** Murky water limits photosynthesis in aquatic plants, disrupting food chains.
  - **Nutrient Overload:** Sediment often carries fertilizers, causing algal blooms and eutrophication.
  - **Damaged Infrastructure:** Siltation reduces reservoir capacity (e.g., 1–2% annual loss in global dam storage).
- **Example:** The Mississippi River carries 550 million metric tons of sediment yearly, creating a "dead zone" in the Gulf of Mexico.

3. Increased Flooding Due to Reduced Water Infiltration- Healthy soil acts like a sponge, absorbing rainwater. Erosion degrades soil structure, reducing infiltration.

- **Impact:**

    - **Higher Surface Runoff:** Water flows rapidly over compacted or bare soil, increasing flood volume and speed.
    - **Flash Floods:** Urban areas and deforested regions experience sudden, destructive floods.
    - **Groundwater Depletion:** Less water seeps into aquifers, worsening droughts.
- **Example:** Deforestation in Haiti has caused catastrophic floods, displacing thousands during hurricanes.

4. Habitat Destruction and Loss of Biodiversity- Erosion removes the soil layer that sustains plants, disrupting entire ecosystems.

- **Impact:**
    - **Plant Loss:** Native vegetation struggles to grow in eroded soil, reducing food and shelter for wildlife.
    - **Species Extinction:** Soil-dependent species (e.g., earthworms, fungi) decline, breaking nutrient cycles.
    - **Fragmented Habitats:** Gullies and landslides split ecosystems, isolating populations.
    - **Coral Reef Damage:** Sediment runoff smothers coral reefs, which host 25% of marine species.
- **Example:** In Madagascar, erosion from slash-and-burn farming has destroyed habitats for lemurs and other endemic species.

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2. **Impact of Air Pollution through Chernobyl Disaster:**

The **Chernobyl disaster**, which occurred on **April 26, 1986**, at the **Chernobyl Nuclear Power Plant** in present-day Ukraine, had severe and long-lasting effects on air pollution. The explosion of Reactor No. 4 released an unprecedented amount of **radioactive materials** into the atmosphere, making it one of the worst nuclear disasters in history.

Impact of Air Pollution through the Chernobyl Disaster

1. Release of Radioactive Contaminants

The explosion resulted in the release of large quantities of **radioactive isotopes** such as:

- **Iodine-131**
- **Cesium-137**
- **Strontium-90**
- **Plutonium-239**

These materials were dispersed into the atmosphere, affecting not just the surrounding area but also **multiple countries across Europe and beyond**.

2. Spread of Radioactive Air Pollution

- The **radioactive cloud** from Chernobyl spread over **Ukraine, Belarus, Russia**, and eventually reached parts of **Europe, Scandinavia, and even North America**.
- Winds carried the radioactive particles, contaminating vast regions, **polluting the air, soil, and water**.

3. Health Effects from Airborne Radiation

- **Acute Radiation Syndrome (ARS):** Many first responders and plant workers suffered from ARS due to direct exposure.
- **Increased Cancer Rates:** Long-term exposure, particularly to **Iodine-131**, significantly increased thyroid cancer cases, especially in children.
- **Respiratory Issues:** Inhalation of radioactive dust caused **lung diseases and other respiratory disorders**.
- **Genetic Mutations:** Prolonged exposure led to **DNA damage**, causing birth defects and other genetic disorders.

4. Environmental Impact

- **Contaminated Air and Soil:** The air pollution caused by radioactive fallout led to the contamination of **forests, rivers, and agricultural lands**.
- **Animal and Plant Mutations:** Many wildlife species experienced **genetic mutations** due to prolonged exposure.
- **Long-Term Radiation Presence:** Elements like **Cesium-137** have long half-lives, meaning affected areas will remain **radioactive for decades**.

5. Global Consequences

- The disaster led to **stricter regulations on nuclear safety** worldwide.
- Airborne radioactive particles even reached **Japan, China, and North America**, proving that nuclear pollution has **global effects**.

3. **Eutrophication:-** Eutrophication is the process by which a water body becomes overly enriched with **nutrients**, leading to excessive growth of algae and other aquatic plants. This process can occur **naturally** over centuries but is often accelerated by **human activities** (known as **cultural eutrophication**).

- **Process:**
    1. Nutrients from agricultural runoff, sewage, and industrial waste enter lakes.
    2. Algae grow rapidly, forming dense blooms.
    3. When algae die, they decompose, consuming oxygen.
    4. Oxygen depletion (hypoxia) harms aquatic life, creating "dead zones.

1. Nutrient Enrichment

- Excess **phosphorus (P) and nitrogen (N)** enter water bodies from sources like:
    - **Agricultural runoff** (fertilizers, pesticides)
    - **Sewage discharge**
    - **Industrial waste**
    - **Detergents** containing phosphates

2. Algal Bloom Formation

- The high concentration of nutrients promotes **rapid algae growth** (algal blooms).
- The algae **cover the water surface**, blocking **sunlight** from reaching underwater plants.

3. Oxygen Depletion (Hypoxia)

- When algae die, **bacteria decompose** them, consuming **large amounts of oxygen**.
- This leads to a **drop in oxygen levels**, creating a condition called **hypoxia**.

4. Death of Aquatic Life

- Low oxygen levels cause fish and other aquatic organisms to **suffocate and die**.
- **Biodiversity decreases**, and only oxygen-tolerant species survive.

5. Formation of "Dead Zones"

- Areas with extremely low oxygen become **"dead zones"**, where most marine life cannot survive

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5. Equipment to Control Air Pollution:

- **Cyclones:** Use centrifugal force to remove particulate matter from gas streams.
- **Electrostatic Precipitators (ESPs):** Use electric charges to capture particles.
- **Scrubbers:** Use liquid to absorb or neutralize pollutants.
- **Bag Filters:** Trap particles in fabric filters.
- **Catalytic Converters:** Convert harmful gases (e.g., NO_x, CO) into less harmful substances.

1. Cyclone collector- Cyclones use **centrifugal force** to separate dust and particulates from gas streams. The polluted air enters the **cyclone chamber**, where spinning motion causes heavier particles to move outward and fall into a collection chamber and Cleaned air exits from the top.

It is **Best For** Removing **larger particulates** (not effective for very fine particles) and Used in industries like **cement plants, sawmills, and metal processing**.

✅ **Advantages:**

- Simple design, **low maintenance**
- Can handle **high temperatures and large volumes** of gas
- No moving parts, making it durable

✅ **Limitations:**

- Less effective for **small particles (<5 microns)**

2) **Electrostatic Precipitators (ESPs)-** ESPs use **high-voltage electric fields** to charge dust particles in polluted air. The charged particles are then attracted to oppositely charged plates, where they stick and are later removed. They are **Best For** Capturing **fine particles (PM2.5, PM10)** from power plants, steel mills, and cement factories.

Advantages:

- **High efficiency** (up to **99% particle removal**)
- Works well for **large volumes of gas**
- Low energy consumption compared to some other methods

Limitations:

- **Expensive installation & maintenance**
- Performance can be affected by **moisture and particle size**

4. **Scrubbers-** Scrubbers use **liquid (usually water or chemical solutions)** to capture or neutralize pollutants. Polluted gases pass through the scrubbing liquid, which absorbs or reacts with harmful substances. The cleaned air is released, and the contaminated liquid is collected for treatment. **It is Best For** Removing **gaseous pollutants** (SO_2, HCl, VOCs) **and** Used in industries like **chemical plants, power plants, and refineries.**

Types of Scrubbers:

- **Wet Scrubbers:** Use water or chemicals to remove **SO_2, NO_x, and particulates**.
- **Dry Scrubbers:** Use dry absorbent materials (like lime) to neutralize acidic gases.

Advantages:

- Effective for **gas and particle removal**
- Can **neutralize toxic gases**
- Reduces **odors**

Limitations:

- **High water usage** (for wet scrubbers)
- Can create **wastewater pollution** that needs treatment

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4. Bag Filters (Fabric Filters)- Contaminated air passes through **fabric filters**, which trap particles while allowing clean air to pass through. The collected dust is periodically removed by shaking or pulse-jet cleaning.**It is Best For** Removing **fine particulates** from industrial processes like **cement, mining, and food production**.

✅ **Advantages:**

- **Very high efficiency** (99% particle removal)
- Works well for **fine dust and fumes**
- Can handle **high dust loads**

✅ **Limitations:**

- **Filters need regular maintenance & replacement**
- Not suitable for **high-temperature gases** unless special materials are used

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5. Catalytic Converters- Found in **vehicles** and **industrial processes**, catalytic converters use a **metal catalyst (platinum, palladium, or rhodium)** to speed up chemical reactions that convert toxic gases into less harmful ones. It is **Best For Vehicles, refineries, and chemical plants**

- **Key Conversions:**
  - **Carbon Monoxide (CO) → Carbon Dioxide (CO_2)**
  - **Nitrogen Oxides (NO_x) → Nitrogen (N_2) and Oxygen (O_2)**
  - **Unburnt Hydrocarbons → Water (H_2O) and CO_2**

Advantages:

- **Reduces toxic gas emissions** significantly
- Helps comply with **air quality regulations**

Limitations:

- **Expensive** (uses rare metals)
- **Can get clogged** over time

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6. Photochemical Smog:

Photochemical smog is a type of air pollution that forms when **sunlight reacts with pollutants**, such as **nitrogen oxides (NO_x) and volatile organic compounds (VOCs)**, to produce harmful secondary pollutants, mainly **ozone (O_3), peroxyacetyl nitrates (PANs), and aldehydes**.

Formation Process:

1. **Emission of Pollutants** – Vehicles, industries, and power plants release **NO_x and VOCs** into the atmosphere.
2. **Sunlight Reaction** – Under sunlight, NO_x and VOCs undergo photochemical reactions, forming harmful secondary pollutants like ozone.
3. **Smog Formation** – The accumulation of **ozone, PANs, and aldehydes** results in a brownish haze called **photochemical smog**.

Common in: Cities with high vehicle emissions and strong sunlight, such as **Los Angeles, Mexico City, Beijing**.

Differences Between Photochemical Smog & London Smog

Feature	Photochemical Smog	London Smog (Sulfurous Smog)
Main Cause	Reaction of NO_x & VOCs with sunlight	High levels of sulfur dioxide (SO_2) from burning coal
Main Pollutants	Ozone (O_3), PANs, NO_2	Sulfur dioxide (SO_2), soot, carbon monoxide (CO)
Appearance	Brownish haze	Thick, grayish-black fog
Weather Conditions	Hot, sunny conditions	Cold, humid conditions
Occurrence	Common in modern urban areas (e.g., Los Angeles, Beijing)	Historically common in industrial cities (e.g., London, 1952 Great Smog)
Effects	Respiratory problems, eye irritation, reduced visibility	Severe respiratory diseases, smothering of cities

Effects of Photochemical Smog

1. Health Effects
- **Respiratory issues** – Worsens asthma, bronchitis, and lung diseases.
- **Eye irritation** – Red, burning eyes due to ozone and aldehydes.
- **Reduced lung function** – Long-term exposure can damage lung tissue.

2. Environmental Effects
- **Damages crops & forests** – Ozone reduces plant growth and photosynthesis.
- **Harms animals** – Toxic pollutants can cause genetic mutations in wildlife.

3. Economic Effects
- **Increased healthcare costs** due to respiratory diseases.
- **Lower crop yields**, affecting food production.
- **Decreased tourism** due to poor air quality and low visibility.

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7. Self-Purification of Rivers:

- **Definition:** The natural ability of rivers to cleanse themselves through physical, chemical, and biological processes.
- **Processes:**
  - **Dilution:** Pollutants are diluted by the river's flow.
  - **Sedimentation:** Heavy particles settle at the bottom.
  - **Oxidation:** Organic matter is broken down by oxygen.
  - **Microbial Decomposition:** Microorganisms degrade organic pollutants.
  - **Sunlight Action:** UV light kills pathogens.

Key Processes of River Self-Purification

1. Dilution (Physical Process)- When pollutants enter a river, they are **diluted** by the flowing water. Faster-moving rivers dilute contaminants more effectively than stagnant water bodies.
Limitation: If the pollution load is too high, dilution alone is insufficient.

2. Sedimentation (Physical Process)-m **Heavier particles** (such as sand, silt, and organic matter) settle to the riverbed due to gravity. Pollutants attached to sediments are **removed from the water column** and trapped in the riverbed.
Example: Heavy metals and phosphorus often settle at the bottom, reducing their impact on water quality.

3. Oxidation & Aeration (Chemical Process)-**Oxygen from air** dissolves into the water, helping break down organic pollutants. **Turbulence and waterfalls** increase oxygen levels (**aeration**), speeding up pollutant breakdown.
Example: The breakdown of organic matter by oxygen leads to the formation of **CO_2 and water**.

4. Biodegradation by Microorganisms (Biological Process)- **Bacteria and fungi** decompose organic waste into simpler, less harmful compounds. **Aerobic bacteria** (require oxygen) break down organic matter into **CO_2, water, and nutrients**. **Anaerobic bacteria** (in low-oxygen conditions) break down waste but produce harmful gases like **methane (CH_4) and hydrogen sulfide (H_2S)**.
Example:
Wastewater rich in organic matter → Bacteria decompose it → Water quality improves.

5. Nitrification & Denitrification (Biological & Chemical Process)-**Nitrification**: Ammonia (NH_3) from organic waste is converted into **nitrates (NO_3^-)** by bacteria. **Denitrification**: In low-oxygen conditions, nitrates are converted into **nitrogen gas (N_2)**, which escapes into the atmosphere.
Example:

- Sewage → Ammonia (NH_3) → Nitrites (NO_2^-) → Nitrates (NO_3^-) → Nitrogen gas (N_2) → Released into the air.

6. Photosynthesis by Aquatic Plants & Algae (Biological Process)-Aquatic plants and algae absorb **carbon dioxide (CO_2)** and produce **oxygen (O_2)** through **photosynthesis**. **Increased oxygen** levels enhance the breakdown of organic matter by bacteria. **Excessive algae growth** (due to pollution) can lead to **eutrophication**, reducing the purification ability of the river.

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8. Noise Pollution:

- **Definition:** Unwanted or harmful sound.
- **Measurement:** Measured in decibels (dB) using a sound level meter.
- **Sources:** Traffic, industrial machinery, construction, and loud music.
- **Effects:** Hearing loss, stress, sleep disturbances, and wildlife disruption.
- **Control Strategies:** Noise barriers, soundproofing, regulations, and public awareness.

1. Definition
Noise pollution refers to **unwanted, excessive, or harmful sound** that disrupts human health, wildlife, and the environment. It is one of the most **widespread environmental pollutants** in urban areas.
Measurement of Noise Pollution- Noise levels are measured in **decibels (dB)** using a **sound level meter**. The human ear can hear sounds between **0 dB (threshold of hearing) to 140 dB (threshold of pain)**.

📊 **Common Noise Levels:**

Noise Source	Decibel Level (dB)	Effect on Health
Whisper	30 dB	No harm
Normal Conversation	60 dB	Comfortable level
Traffic Noise	70-85 dB	Annoying but tolerable
Construction Work	90-110 dB	Hearing damage with long exposure
Loud Music/Concert	110-120 dB	Potential hearing loss
Jet Engine (Close)	130-140 dB	Immediate pain and damage

3. Sources of Noise Pollution
Human Activities

- **Traffic Noise** – Cars, motorcycles, honking, and sirens.
- **Industrial Machinery** – Factories, power plants, and metalworking.
- **Construction Noise** – Drilling, hammering, and heavy equipment.
- **Loud Music & Entertainment** – Concerts, nightclubs, and home music systems.

Environmental Sources

- **Thunderstorms and earthquakes**

- **Airplane and train noise**
- **Animal sounds (e.g., barking dogs, bird calls in urban areas)**

4. Effects of Noise Pollution

🚑 **Health Effects**

- **Hearing Loss** – Prolonged exposure to noise above **85 dB** can cause permanent damage.
- **Increased Stress & Anxiety** – High noise levels trigger **stress hormones**, leading to mental fatigue.
- **Sleep Disturbances** – Loud environments reduce sleep quality, causing **insomnia and fatigue**.
- **Cardiovascular Issues** – Long-term exposure can **increase blood pressure and risk of heart disease**.

🐾 **Effects on Wildlife**

- **Interferes with communication** – Birds, dolphins, and other animals rely on sound for communication.
- **Disrupts migration & reproduction** – Marine animals, like whales, suffer from underwater noise pollution.
- **Increases predation risk** – Some animals struggle to detect predators due to noise interference.

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5. Noise Pollution Control Strategies

Engineering Controls

- **Noise Barriers** – Walls and green belts along highways absorb sound.
- **Soundproofing Buildings** – Using acoustic panels, double-glazed windows, and insulation.
- **Quiet Machinery Design** – Developing low-noise engines and equipment.

Regulatory Measures

- **Imposing Noise Limits** – Governments set **permissible noise levels** in different zones.
- **Banning Unnecessary Honking** – Strict fines for honking in silent zones (e.g., near hospitals & schools).
- **Time Restrictions** – Limiting loud activities (e.g., fireworks and construction) to certain hours.

Public Awareness & Behavioral Changes

- **Encouraging use of silent zones** (libraries, hospitals, residential areas).
- **Promoting responsible use of loudspeakers and music systems**.
- **Using noise-friendly urban planning** (planting trees, zoning industries away from homes)

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9. On-Site and Off-Site Disposal Techniques:

- **On-Site:** Composting, septic tanks, and pit latrines.
- **Off-Site:** Landfills, incineration, and centralized sewage treatment plants.

a) **Source Reduction & Waste Minimization- Reducing waste at the source** by using eco-friendly materials.

- **Examples:**
  - Using reusable bags instead of plastic.
  - Avoiding excessive packaging.
  - Promoting minimal waste production in industries.

b) **Composting (For Organic Waste)-** Converts **biodegradable waste** into compost, enriching soil fertility.

- **Methods:**
  - **Aerobic composting** – With oxygen (faster).
  - **Anaerobic composting** – Without oxygen (produces methane, used for biogas).
  - **Vermicomposting** – Using earthworms to break down waste

c) **Recycling & Reuse - Reprocessing materials** like plastic, paper, metal, and glass into new products.

- **Example:**
  - Recycling **plastic bottles** into synthetic fibers.
  - Reusing **old furniture** by refurbishing.

d) **Waste-to-Energy (WTE) Conversion-** Generating energy from solid waste through **thermal and biological processes**.

- **Techniques:**
  - **Incineration** – Burning waste at high temperatures to produce heat & electricity.

- **Pyrolysis** – Heating waste in the absence of oxygen to produce biofuel.
- **Biogas Production** – Anaerobic digestion of organic waste to generate methane.

e) **Landfilling (Final Disposal Method)**

- **Used for non-recyclable and residual waste**.
- Modern **sanitary landfills** use **liners and gas collection systems** to prevent pollution.
- **Issues:**
    - Requires large land areas.
    - Can cause groundwater contamination if not managed properly.

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10. Good Ozone and Bad Ozone:

- **Good Ozone:** Found in the stratosphere, it protects Earth from UV radiation.
- **Bad Ozone:** Found at ground level, it is a harmful component of smog.
- **Formation:** Good ozone forms naturally in the stratosphere, while bad ozone forms from NO_x and VOCs reacting in sunlight.

1. **Good Ozone (Stratospheric Ozone)**

Location: Found in the **stratosphere (10-50 km above Earth)**.
Role: Forms the **ozone layer**, which **absorbs harmful ultraviolet (UV) radiation** from the Sun.
Formation:

- Oxygen molecules (**O_2**) absorb high-energy **UV radiation**, splitting into single oxygen atoms (**O**).
- These single atoms react with O_2 to form **O_3 (ozone)**.

Without the ozone layer, life on Earth would be exposed to dangerous UV radiation, leading to increased risks of skin cancer, cataracts, and DNA damage.
Threat to Good Ozone:

- Human activities release **chlorofluorocarbons (CFCs)** from aerosols and refrigerants.
- CFCs break down ozone molecules, causing the **ozone hole** (especially over Antarctica).

2. Bad Ozone (Tropospheric Ozone)

Location: Found at **ground level in the troposphere**.
Role: A major component of **photochemical smog**, harmful to health and the environment.
Formation:

- Emissions of **nitrogen oxides (NO_x) and volatile organic compounds (VOCs)** from vehicles, industries, and power plants react with **sunlight** to produce ozone.

🔹 **Unlike stratospheric ozone, which protects life, tropospheric ozone is a pollutant that causes respiratory problems, damages crops, and contributes to climate change.**

🚨 **Effects of Bad Ozone:**

- **Health Issues**: Causes lung irritation, asthma, and reduced lung function.
- **Environmental Damage**: Reduces crop yields and harms ecosystems.
- **Climate Impact**: Acts as a **greenhouse gas**, trapping heat in the atmosphere.

Feature	Good Ozone (Stratospheric)	Bad Ozone (Tropospheric)
Location	Stratosphere (10-50 km above Earth)	Troposphere (ground level)
Formation	Natural UV reactions with oxygen	Pollution from NO_x & VOCs reacting with sunlight
Role	Protects from UV radiation	Major air pollutant in smog
Effects	Essential for life	Causes respiratory diseases, harms plants & climate

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11. Solid Waste Management:

Solid waste is a non-liquid, non-soluble material ranging from municipal garbage to industrial waste that sometimes contains complex and hazardous substances. It includes domestic waste, sanitary waste, commercial waste, institutional waste, catering and market waste, bio-medical waste, and e-waste.

Types of Solid Waste Management

- **Landfill:** It involves burying the waste in vacant locations around the city. The dumping site should be covered with soil to prevent contamination.
  Benefits: A sanitary disposal method if managed effectively.
  Limitations: A reasonably large area is required.
- **Incineration:** It is the controlled oxidation (burning/thermal treatment) of mostly organic compounds at high temperatures to produce thermal energy, CO_2, and water.
  Benefits: Burning significantly reduces the volume of combustible waste.
  Limitations: Smoke and fire hazards may exist.
- **Composting:** It is a natural process of recycling organic matter like leaves and food scraps into beneficial fertilizers that can benefit both soil and plants.
  Benefits: It is beneficial for crops and is an environment-friendly method.
  Limitations: Requires high-skilled labour for large-scale operation.
- **Recycling:** It is a process of converting waste material into new material. Examples: wood recycling, paper recycling, and glass recycling.
  Benefits: It is environment-friendly.
  Limitations: It is expensive to set up and not reliable in case of an emergency.
- **Vermicomposting:** Vermicomposting is a bio-conversion technique that is commonly used to handle solid waste. Earthworms feed on organic waste to reproduce and multiply in number, vermicompost, and vermiwash as products in this bio-conversion process.
  Benefits: It reduces the need for chemical fertilizers and enhances plant growth.
  Limitations: It is time-consuming, cost-ineffective, and requires extra care.

Aspect	**Incineration**	**Combustion**
Definition	Complete burning of waste at high temperatures to reduce it to ash.	General process of burning materials for heat or energy.
Purpose	Waste reduction & energy recovery.	Primarily for energy production.
Temperature	850-1200°C	500-800°C
End Products	Ash, flue gases, and energy.	Heat, gases, and residue.
Pollution Control	Requires advanced filters to reduce emissions.	May produce more emissions if not controlled properly.
Example	Municipal solid waste incinerators.	Coal, biomass, or waste used in power plants.

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12. Classification of Wastes:

Waste can be classified into different categories based on their **composition, properties, and impact on the environment**. Proper classification is crucial for effective waste management. Here are the main types of waste:

1. By Composition

A) Biodegradable Waste- Waste that can be decomposed by natural processes (e.g., bacteria and fungi).

- **Examples:** Food scraps, garden waste, paper, wood.
- **Management Methods:** Composting, biogas production, vermicomposting.

b. Non-Biodegradable Waste

- **Definition:** Waste that does not break down naturally or decomposes very slowly.
- **Examples:** Plastics, glass, metals, ceramics.
- **Management Methods:** Recycling, landfilling, waste-to-energy.

c. Hazardous Waste

- **Definition:** Waste that poses a risk to human health or the environment due to its chemical or biological properties.
- **Examples:** Batteries, pesticides, chemicals, medical waste, asbestos.

- **Management Methods:** Specialized treatment, safe disposal, incineration.

d. E-Waste

- **Definition:** Waste from electronic products that are no longer in use or have reached the end of their lifecycle.
- **Examples:** Old computers, mobile phones, televisions, batteries.
- **Management Methods:** Recycling, recovery of valuable metals (gold, silver), safe disposal.

2. By Origin

a. Industrial Waste

- **Definition:** Waste generated by manufacturing processes or industrial operations.
- **Examples:** Scrap metal, chemicals, sludge, packaging.
- **Management Methods:** Recycling, treatment, and disposal in landfills or incinerators.

b. Agricultural Waste

- **Definition:** Waste produced in farming activities, such as crop residue, animal manure, and pesticides.
- **Examples:** Crop stalks, manure, pesticides, fertilizers.
- **Management Methods:** Composting, biogas production, and responsible disposal.

c. Household Waste

- **Definition:** Waste generated in residential areas from daily living activities.
- **Examples:** Kitchen waste, packaging materials, old clothes, electronics.
- **Management Methods:** Sorting, recycling, composting, and responsible disposal.

3. By Physical Form

a. Solid Waste

- **Definition:** Waste in solid form that does not flow like liquids or gases.
- **Examples:** Paper, plastic, glass, metals.
- **Management Methods:** Recycling, landfill, incineration.

b. Liquid Waste

- **Definition:** Waste in the form of liquids, often requiring special treatment.
- **Examples:** Wastewater from industries, sewage, cleaning agents, oils.
- **Management Methods:** Treatment plants, filtration, and chemical neutralization.

c. Gaseous Waste

- **Definition:** Waste in the form of gases, typically produced in industrial processes or during combustion.
- **Examples:** Carbon dioxide, methane, sulfur dioxide, nitrogen oxides.
- **Management Methods:** Emission control technologies, filtering systems, and carbon capture.

4. By Decomposability

a. Organic Waste

Definition: *Waste composed of carbon-based materials that can be decomposed by microorganisms.*

- **Examples:** Food waste, plant matter, and animal waste.
- **Management Methods:** Composting, anaerobic digestion, biogas production.

b. Inorganic Waste

- **Definition:** Waste that consists of non-carbon-based materials that do not decompose easily.
- **Examples:** Metals, plastics, glass, and ceramics.
- **Management Methods:** Recycling, incineration, landfill.

5. By Impact on Environment

a. Recyclable Waste

- **Definition:** Waste that can be reprocessed into new materials or products.
- **Examples:** Paper, plastic, metal, glass.
- **Management Methods:** Recycling, reprocessing.

b. Non-Recyclable Waste

- **Definition:** Waste that cannot be reused or recycled, and must be disposed of.
- **Examples:** Contaminated food packaging, broken ceramics, certain plastics.

- **Management Methods:** Landfill, incineration.

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6. By Toxicity

a. Non-Toxic Waste

- **Definition:** Waste that does not harm human health or the environment.
- **Examples:** Non-contaminated food waste, paper, and cardboard.
- **Management Methods:** Composting, recycling.

b. Toxic Waste

- **Definition:** Waste that contains harmful chemicals that can pose risks to humans and the environment.
- **Examples:** Pesticides, heavy metals, industrial solvents.
- **Management Methods:** Special disposal, chemical treatment, and secure storage.

13. Air Pollution:

Air pollution refers to the **presence of harmful substances** in the atmosphere that degrade air quality and pose risks to human health, the environment, and the climate. These pollutants can be in the form of gases, particles, or biological molecules that interfere with the natural composition of the atmosphere.

Sources of Air Pollution

1. **Natural Sources**:
   - **Volcanic eruptions** (releasing sulfur dioxide and ash).
   - **Forest fires** (producing particulate matter and carbon monoxide).
   - **Dust storms** and **pollen**.
2. **Human-made Sources**:
   - **Vehicle emissions** (CO, NO_x, particulate matter).
   - **Industrial activities** (SO_2, VOCs, NO_x).
   - **Burning of fossil fuels** (coal, oil, natural gas).
   - **Agricultural activities** (methane, ammonia).
   - **Household activities** (use of chemicals, cooking).

Effects of Air Pollution on Human Health

1. Respiratory and Cardiovascular Problems

- **Asthma**: Inhalation of particulate matter (PM) and gases (e.g., NO_2) can aggravate asthma symptoms.
- **Chronic Respiratory Diseases**: Long-term exposure to pollutants like **smog, sulfur dioxide (SO_2), and particulate matter** can cause chronic bronchitis and emphysema.
- **Heart Disease**: Pollutants like **carbon monoxide (CO)** and **ozone (O_3)** can damage blood vessels, increase heart disease risk, and cause high blood pressure.

2. Cancer

- **Lung Cancer**: Prolonged exposure to **carcinogenic air pollutants** such as **benzene**, **formaldehyde**, and **radon** increases the risk of lung cancer.
- **Other cancers**: Studies suggest air pollution may also increase the risk of cancers in organs like the bladder, kidney, and liver.

3. Neurological Impacts

- **Cognitive Decline**: Air pollution has been linked to developmental issues and cognitive decline, especially in children and the elderly.
- **Stroke**: Long-term exposure to pollutants like particulate matter can increase the risk of stroke by affecting the blood vessels and brain.

4. Effects on Reproductive Health

- **Fertility Issues**: Some pollutants like **lead** and **phthalates** can impact reproductive health, leading to fertility problems in men and women.
- **Prenatal Exposure**: Exposure to air pollution during pregnancy can lead to premature birth, low birth weight, and developmental issues in newborns.

5. Premature Death

- Exposure to hazardous air pollutants is responsible for millions of premature deaths globally each year, especially from diseases such as lung cancer, respiratory infections, and heart disease.

Control Measures for Air Pollution

1. Regulatory Measures

- **Setting Emission Standards**: Governments establish **emission limits** for vehicles, industries, and power plants to reduce the release of harmful pollutants.
- **Air Quality Monitoring**: Installing **monitoring stations** to assess air quality and take action when pollutant levels exceed safe limits.
- **Enforcing Clean Air Acts**: Laws and regulations such as the **Clean Air Act** in the U.S. that set standards for air pollutants, including sulfur dioxide (SO_2), nitrogen oxides (NO_x), and particulate matter.

2. Technological Measures

- **Filters & Scrubbers**: **Industrial scrubbers** and **electrostatic precipitators** capture particulate matter and harmful gases like SO_2, NO_x, and VOCs before they enter the atmosphere.
- **Catalytic Converters**: Installed in **vehicles** to reduce emissions of harmful gases like carbon monoxide (CO), nitrogen oxides (NO_x), and hydrocarbons.
- **Clean Energy Technologies**: Promoting **renewable energy** sources like solar, wind, and hydropower to reduce reliance on fossil fuels.

3. Vehicle and Traffic Control

- **Public Transportation Systems**: Encouraging the use of **electric buses, trains, and carpooling** to reduce traffic emissions.
- **Vehicle Emission Standards**: Strict **vehicle emission regulations** and the promotion of **electric vehicles (EVs)** to reduce air pollutants.
- **Traffic Management**: Implementing **congestion charges, car-free zones**, and promoting cycling and walking.

4. Industrial Pollution Control

- **Green Technologies**: Encouraging the adoption of **eco-friendly technologies** and processes that emit fewer pollutants.
- **Energy Efficiency**: Improving energy efficiency in industries to reduce overall emissions and the consumption of fossil fuels.

5. Awareness and Education

- **Public Awareness Campaigns**: Educating people about the **health risks** of air pollution and how to reduce their exposure (e.g., avoiding outdoor activities during smog).
- **Promoting Green Practices**: Encouraging individuals to use energy-efficient appliances, reduce waste, and adopt eco-friendly lifestyle choices.

6. Urban Planning

- **Greening Cities**: Planting trees and creating green spaces in urban areas to absorb pollutants and provide clean air.
- **Zoning and Waste Management**: Ensuring proper waste disposal, industrial zoning away from residential areas, and reducing open burning of waste.

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14. Primary and Secondary Pollutants:

- **Primary Pollutants:** Emitted directly (e.g., CO, SO_2, NO_x).
- **Secondary Pollutants:** Formed by reactions in the atmosphere (e.g., ozone, PAN).
- **Effects:** Primary pollutants cause direct harm, while secondary pollutants contribute to smog and acid rain.

Primary pollutants	Secondary pollutants
Primary pollutants are emitted directly from the source.	Secondary pollutants are not emitted directly from the source but are formed due to chemical reactions.
They are found in the atmosphere in the form they are emitted.	They are found as products of chemical reaction between the atmospheric constituents and primary pollutants.
Ash, smoke, dust, oxides of carbon, sulphur and nitrogen are primary pollutants.	SO_3, O_3, ketones and hydrogen cyanide are secondary pollutants.

The effect of primary pollutants is both direct and indirect.
Primary pollutants are emitted by a source directly into the atmosphere, affecting all living organisms.
They also undergo a chemical reaction, forming secondary pollutants and indirectly affecting the ecosystem. For example, the formation of sulphur dioxide is a direct effect of primary pollutants.
Whereas sulphur dioxide reacting with water to form sulphuric acid, which is responsible for acid rain, is an indirect effect. The two main sources of these pollutants are natural and man-made.

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15. Water Pollution:

- **Definition:** Contamination of water bodies by harmful substances.
- **Sources:** Industrial discharge, agricultural runoff, and sewage.
- **Treatment Steps:** Screening, sedimentation, filtration, disinfection, and advanced treatment.

Water pollution can be defined as the contamination of water bodies. Water pollution is caused when water bodies such as rivers, lakes, oceans, groundwater and aquifers get contaminated with industrial and agricultural effluents.
When water gets polluted, it adversely affects all lifeforms that directly or indirectly depend on this source. The effects of water contamination can be felt for years to come.

Sources Of Water Pollution
The key causative of water pollution in India are:

- Urbanization.
- Deforestation.
- Industrial effluents.
- Social and Religious Practices.
- Use of Detergents and Fertilizers.
- Agricultural run-offs- Use of insecticides and pesticides.

Unit 4

Very Short Questions

Q1. What is El-Nino?

It is a warming of the ocean surface, or above-average sea surface temperatures, in the central and eastern tropical Pacific Ocean.

Q2. Differentiate climate and weather.

The difference between weather and climate is a measure of time. Weather refers to the short-term, usually 24-hour atmospheric conditions of a particular location. Climate refers to the average atmospheric conditions for a relatively long period of time, usually 30 years.

Q3. Define GRAP.

The Graded Response Action Plan (GRAP) is a set of guidelines and measures implemented to combat air pollution in India's National Capital Region (NCR), which includes Delhi and its surrounding areas.

Q4. What do you mean by kyoto and montreal protocol?

Kyoto Protocol:

The Kyoto Protocol was an international treaty which extended the 1992 United Nations Framework Convention on Climate Change (UNFCCC) that commits state parties to reduce greenhouse gas emissions, based on the scientific consensus that global warming is occurring and that human-made CO2 emissions are driving it. The Kyoto Protocol was adopted in Kyoto, Japan, on 11 December 1997 and entered into force on 16 February 2005. There were 192 parties (Canada withdrew from the protocol, effective December 2012) to the Protocol in 2020.

The Kyoto Protocol applied to the seven greenhouse gases listed in Annex A: carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF6), nitrogen trifluoride (NF3). Nitrogen trifluoride was added for the second compliance period during the Doha Round.

Montreal Protocol:

The Montreal Protocol, finalized in 1987, is a global agreement to protect the stratospheric ozone layer by phasing out the production and consumption of ozone-depleting substances (ODS). ODS are substances that were commonly used in products such as refrigerators, air conditioners, fire extinguishers, and aerosols.

Q5. Does depletion of ozone layer increase ground level ultraviolet radiation?

The depletion of the ozone layer leads, on the average, to an increase in ground-level ultraviolet radiation, because ozone is an effective absorber of ultraviolet radiation. The Sun emits radiation over a wide range of energies, with about 2% in the form of high-energy, ultraviolet (UV) radiation. Some of this UV radiation (UV-B) is especially effective in causing damage to living beings, for example, sunburn, skin cancer, and eye damage to humans. The amount of solar UV radiation received at any particular location on the Earth's surface depends upon the position of the Sun above the horizon, the amount of ozone in the atmosphere, and local cloudiness and pollution. Scientists agree that, in the absence of changes in clouds or pollution, decreases in atmospheric ozone lead to increases in ground-level UV radiation.

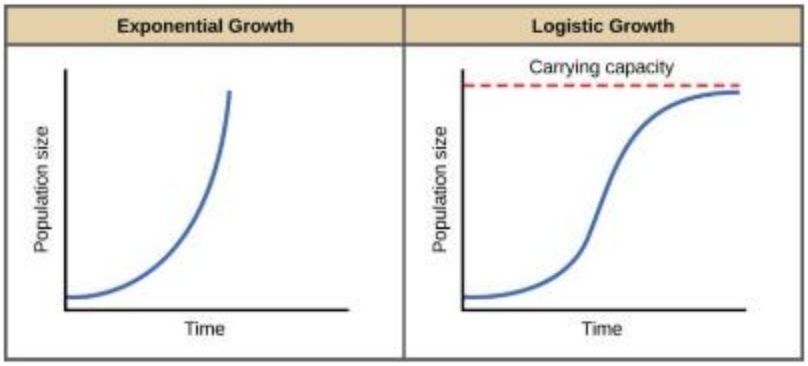
Q6. What is the difference between climate change and climate variability?

Climate variability includes all the variations in the climate that last longer than individual weather events, whereas the term climate change only refers to those variations that persist for a longer period of time, typically decades or more.

Q7. Differentiate between J-shaped and S- shaped population growth curve.

S-shaped pattern of population growth form shows an initial gradual increase followed by an exponential increase and then a gradual decline to a near constant level. It shows logistic growth model which defines the concept of ‘survival of the fittest’. Thus, it considers the fact that resources in nature are exhaustible. The term ‘Carrying capacity’ defines the limit of the resources beyond which they cannot support any number of organisms. As the population reaches the carrying capacity, the curve begins to take an “S” shape. Logistical growth is seen in all stable populations living in a finite geographic area.

J-shaped pattern shows exponential population growth and its abrupt crash after attaining the peak value. In an ideal condition where there is an unlimited supply of food and resources, the population growth will follow an exponential order. There is no competition or limit to the exponential growth. The population starts small and grows rapidly as time progresses, giving a J-type exponential growth curve. Exponential growth usually occurs in regions that have newly been colonized.



Q8. Define urbanisation.
Urbanisation is the increase in the proportion of people living in towns and cities. Urbanisation occurs because people move from rural areas (countryside) to urban areas (towns and cities) and it results in growth in the size of the urban population and the extent of urban areas.

Long Questions
Q1. Discuss the phenomenon of 'greenhouse effect'? Name four greenhouse gases. Discuss the causes, effects and remedies of greenhouse effect.
The Natural Greenhouse Effect
The greenhouse effect is the rise in temperature that the Earth experiences because certain gases in the atmosphere (water vapor, carbon dioxide, nitrous oxide, ozone, methane, for example) trap energy that comes from the sun. These gases are usually called greenhouse gases since they behave much like the glass panes in a greenhouse. The glass panels of the greenhouse let in the light but keep heat from escaping and this is similar to the effect these gasses have on earth. Sunlight enters the Earth's atmosphere, passing through the greenhouse gases. As it reaches the Earth's surface, land, water, and biosphere absorb the sunlight's energy. Once absorbed, this energy is sent back into the atmosphere. Some of the energy passes back into space, but much of it remains trapped in the atmosphere by the greenhouse gases. This is the completely natural and very important process, because without the greenhouse effect, the Earth would not be warm enough for humans to live.

The Enhanced Greenhouse Effect Some human activities such as burning fossil fuels - coal, oil and natural gas - releases carbon dioxide into the atmosphere. Cutting down and burning trees also produces a lot of carbon dioxide. A group of greenhouse gases called the chlorofluorocarbons have been used in aerosols, such as hairspray cans, fridges and in making foam plastics. Since there are more and more greenhouse gases in the atmosphere, more heat is trapped, which makes the Earth warmer. This is known as **global warming**.

The major greenhouse gases are carbon dioxide, ozone, methane, nitrous oxide, chlorofluorocarbons (CFCs) and water vapours.
Causes of greenhouse effect:
Deforestation: Trees help to keep the climate by **absorbing carbon dioxide** from the atmosphere. When we cut down trees, that beneficial effect is lost and the carbon stored in the trees is released.

Burning of fossil fuels: The burning of fossil fuels such as natural gas, coal and oil produce more **carbon dioxide**.

Agricultural activity: Cows and sheep produce a large quantity of **methane** when they digest their food. Fertilisers have nitrogen, which produces **nitrous oxide**. Fertiliser is a natural/chemical substance added to land or soil for the better growth of plants.

Decay of dead organism: After the death of an organism, the decomposition releases **carbon** into the air, soil and water, resulting in **carbon dioxide** formation.

Thermal power plant: Thermal power plants pump out many greenhouse gases as by-products while burning fossil fuels. **Carbon dioxide** is one of the main gases.

Pollution of an automobile: Gas consuming car and truck emits **carbon dioxide** and other greenhouse gases.

Eruption of volcanoes: When **volcanoes erupt**, they emit a mixture of **gases** along with **ashes** and **particles**. Volcanic gases like sulfur dioxide can cause **global cooling**, while **carbon dioxide** causes global warming.

Industrialisation: The rapid population growth associated with industrialisation because a **larger population** leads to increased demand for products, higher production levels, consumption and automatically rises levels of **greenhouse gases**.

Effects of greenhouse effect:

- With an increase in the temperature the **average temperature increases**.
- Ice melts at a rapid pace.
- Rise in the **sea levels**.
- Leads to **soil erosion** and **unseasonal rains**.
- Increase in spreading of waterborne and insect-borne diseases.
- As weather and temperature changes, the plants and animals will be affected all over the world. For example, polar bears and seals will have to find new land for hunting and living if the ice in the Arctic melts.
- It may lead to increased risks of malnutrition, coastal flooding, huge population displacement, and new diseases moving into the regions which lack either population immunity or a strong public health infrastructure.

Prevention of Greenhouse Effect

Now that we have made a list of causes, finding alternatives to these causes becomes by following the below-mentioned preventive measures:

- **Afforestation:** Afforestation on a large scale area helps in decreasing the release of carbon dioxide in the atmosphere.
- **Conservation of energy:** Switching to renewable sources of energy such as solar energy, wind energy, etc will reduce the use of fossil fuels. This eventually reduces the release of carbon dioxide into the atmosphere.
- **Policy intervention:** When the government comes up with strict policies to maintain the overall air quality of the city.

Q2. Discuss the phenomena of Global warming along with its causes, effects and control measures.

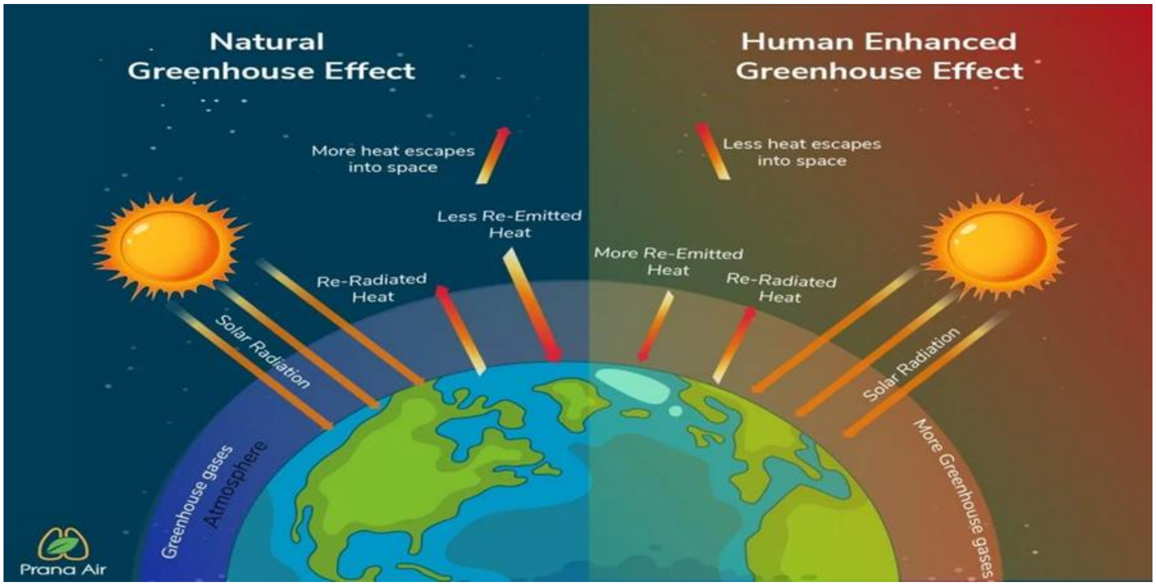
Global Warming

• Troposphere, the lowermost layer of the atmosphere, traps heat by a natural process due to the presence of certain gases. This effect is called Green House Effect as it similar to the warming effect observed in the horticultural greenhouse made of glass.

• The amount of heat trapped in the atmosphere depends mostly on the concentrations of “heat trapping” or “greenhouse” gases and the length of time they stay in the atmosphere. The major greenhouse gases are carbon dioxide, ozone, methane, nitrous oxide, chlorofluorocarbons (CFCs) and water vapours. Heat trapped by greenhouse gases in the atmosphere keeps the planet warm, at the global scale it is called global warming.

What are the Greenhouse Gases?

A greenhouse gas absorbs and emits radiant energy in the thermal infrared region, resulting in the greenhouse effect. Greenhouse gases are essential for maintaining a livable temperature on Earth. The average surface temperature of the Earth would be around 18°C if not for greenhouse gases. Greenhouse gases keep temperatures high in the lower atmosphere, enabling less heat to escape. CO2 is the most released greenhouse gas, despite the fact that water vapor is the most prevalent greenhouse gas naturally existing in the atmosphere.



Causes of global warming:

The greenhouse gases occur naturally; however, **human activities increase the concentration** of the gases. Some of these activities are listed below:

- Deforestation
- Burning fossil fuels
- Agricultural activity
- Decay of dead organism
- Thermal power plant
- Pollution of automobile
- Eruption of volcanoes
- Industrialization

Deforestation: Trees help to keep the climate by **absorbing carbon dioxide** from the atmosphere. When we cut down trees, that beneficial effect is lost and the carbon stored in the trees is released.

Burning of fossil fuels: The burning of fossil fuels such as natural gas, coal and oil produce more **carbon dioxide**.

Agricultural activity: Cows and sheep produce a large quantity of **methane** when they digest their food. Fertilisers have nitrogen, which produces **nitrous oxide**. Fertiliser is a natural/chemical substance added to land or soil for the better growth of plants.

Decay of dead organism: After the death of an organism, the decomposition releases **carbon** into the air, soil and water, resulting in **carbon dioxide** formation.

Thermal power plant: Thermal power plants pump out many greenhouse gases as by-products while burning fossil fuels. **Carbon dioxide** is one of the main gases.

Pollution of an automobile: Gas consuming car and truck emits **carbon dioxide** and other greenhouse gases.

Eruption of volcanoes: When **volcanoes erupt**, they emit a mixture of **gases** along with **ashes** and **particles**. Volcanic gases like sulfur dioxide can cause **global cooling**, while **carbon dioxide** causes global warming.

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- It may lead to increased risks of malnutrition, coastal flooding, huge population displacement, and new diseases moving into the regions which lack either population immunity or a strong public health infrastructure.

How to prevent global warming?

- Minimise the use of **fossil fuels**.
- Plant more **trees**.
- Restrict the use of **CFC**s (chloro fluorocarbon).
- Use a product that can be reusable and recycling.
- Buy an energy-efficient product, which burns **less fuel** for making **electricity**.
- Afforestation: Afforestation on a large scale area helps in decreasing the release of carbon dioxide in the atmosphere.
- Conservation of energy: Switching to renewable sources of energy such as solar energy, wind energy, etc will reduce the use of fossil fuels. This eventually reduces the release of carbon dioxide into the atmosphere.
- Policy intervention: When the government comes up with strict policies to maintain the overall air quality of the city.

Q3. What is ozone hole? Discuss the reactions involved in the formation and depletion of ozone layer. Also explain the causes, effects and control of ozone layer depletion.

Solution: For the last 450 million years the earth has had a natural sunscreen in the stratosphere called the ozone layer. This layer filters out harmful ultraviolet radiations from the sunlight and thus protects various life forms on the earth. Ozone is a form of oxygen. The molecule of oxygen contains two atoms whereas that of ozone contains three (O_3). In the stratosphere ozone is continuously being created by the absorption of short wave-length ultraviolet (UV) radiations. Ultraviolet radiations less than 242 nanometres decompose molecular oxygen into atomic oxygen (O) by photolytic decomposition.

$O_2 + hv \rightarrow O + O$ The atomic oxygen rapidly reacts with molecular oxygen to form ozone.
$O + O_2 + M \rightarrow O_3 + M$

(M is a third body necessary to carry away the energy released in the reaction). Ozone thus formed distributes itself in the stratosphere and absorbs harmful ultraviolet radiations (200 to 320 nm) and is continuously being converted back to molecular oxygen. $O_3 + hv \rightarrow O_2 + O$. Absorption of UV radiations results in heating of the stratosphere. The net result of the above reactions is an equilibrium concentration of ozone. This equilibrium is disturbed by reactive atoms of chlorine, bromine etc. which destroy ozone molecules and result is thinning of ozone layer generally called **ozone hole**. Ozone hole is a severe area of ozone depletion that occurs in stratosphere over the polar regions. It is characterized by a large area of extremely low ozone concentrations.

When the chlorine and bromine atoms in the atmosphere come in contact with ozone, it destroy the ozone molecules. One chlorine can destroy 100,000 molecules of ozone. It is destroyed more quickly than it is created.

$CFCI_3$ + UV Light = => $CFCL_2$ +Cl
Cl + O_3 = => ClO +O_2
ClO + O = = > Cl + O_2
The free chlorine atom is then free to attack another ozone molecule.

$Cl + O_3 ==> ClO + O_2$
$ClO + O ==> Cl + O_2$
And again
$Cl + O_3 ==> ClO + O_2$
$ClO + O ==> Cl + O_2$
And again… for thousands of times.

Causes for ozone layer Depletion

Natural Causes
The ozone layer has been found to be depleted by certain natural processes such as Sun-spots and stratospheric winds. But it does not cause more than 1-2% of the ozone layer depletion.
The volcanic eruptions are also responsible for the depletion of the ozone layer.

Man-made Causes of Depletion of the Ozone Layer
Chlorofluorocarbons
Chlorofluorocarbons or CFCs are the main cause of ozone layer depletion. These are released by solvents, spray aerosols, refrigerators, air-conditioners, etc.
The molecules of chlorofluorocarbons in the stratosphere are broken down by ultraviolet radiations and release chlorine atoms. These atoms react with ozone and destroy it.

Unregulated Rocket Launches
Researches say that the unregulated launching of rockets results in much more depletion of the ozone layer than the CFCs do. If not controlled, this might result in a huge loss of the ozone layer by the year 2050.

Nitrogenous Compounds
The nitrogenous compounds such as NO_2, NO, N_2O are highly responsible for the depletion of the ozone layer.

Effects

• The UV-B radiations affect DNA and the photosynthetic chemicals. Any change in DNA can result in mutation and cancer. Cases of skin cancer (basal and squamous cell carcinoma) which do not cause death, but cause disfigurement will increase.
• Easy absorption of UV rays by the lens and cornea of eye will result in increase in incidents of cataract.
• Melanin producing cells of the epidermis (important for hu-man immune system) will be destroyed by UV-rays resulting in immuno-suppression. Fair people (can't produce enough melanin) will be at a greater risk of UV exposure.
• Phytoplanktons are sensitive to UV exposure. Ozone depletion will result in decrease in their population thereby affecting the population of zooplankton, fish, marine animals, in fact the whole aquatic food chain.
• Direct exposure to ultraviolet radiations leads to skin and eye cancer in animals.

• Yield of vital crops like corn, rice, soybean, cotton, bean, pea, sorghum and wheat will decrease. Strong ultraviolet rays may lead to minimal growth, flowering and photosynthesis in plants. The forests also have to bear the harmful effects of the ultraviolet rays.
• Degradation of paints, plastics and other polymer material will result in economic loss due to effects of UV radiation resulting from ozone depletion.

Solutions to Ozone Layer Depletion

The depletion of the ozone layer is a serious issue and various programmes have been launched by the government of various countries to prevent it. However, steps should be taken at the individual level as well to prevent the depletion of the ozone layer. Following are some points that would help in preventing this problem at a global level:

Avoid Using ODS
Reduce the use of ozone depleting substances. E.g. avoid the use of CFCs in refrigerators and air conditioners, replacing the halon based fire extinguishers, etc.

Minimise the Use of Vehicles
The vehicles emit a large amount of greenhouse gases that lead to global warming as well as ozone depletion. Therefore, the use of vehicles should be minimised as much as possible.

Use Eco-friendly Cleaning Products
Most of the cleaning products have chlorine and bromine releasing chemicals that find a way into the atmosphere and affect the ozone layer. These should be substituted with natural products to protect the environment.

Use of Nitrous Oxide should be Prohibited
The government should take actions and prohibit the use of harmful nitrous oxide that is adversely affecting the ozone layer. People should be made aware of the harmful effects of nitrous oxide and the products emitting the gas so that its use is minimised at the individual level as well.

Less use of pesticides: pesticides helps in growing your farms and plants but cause harm to the ozone layer and contribute to ozone layer depletion.

Limited use of harmful chemicals for cleaning: the chemicals used for making cleaning products results in depletion of the ozone layer.

Q4. How is domestic discharge responsible for increasing the BOD level into the stream?

Sewage discharge contains organic matter, when the sewage is discharged into a water body, the bacteria present in the water decompose the organic matter in the sewage and for this, large amount of dissolved oxygen is consumed by the bacteria. This results in the increase of BOD. Increase in BOD affects the aquatic life like aquatic plants and fishes dwelling in that water body since they will face lack of dissolved oxygen.

Domestic sewage is the waste originating from the kitchen, toilet, laundry, and other sources. It contains impurities such as suspended solid (sand, salt, clay), colloidal material (fecal matter, bacteria, plastic and cloth fiber), dissolved materials (nitrate, phosphate, calcium, sodium, ammonia), and disease-causing microbes. When organic wastes from the sewage enter the water bodies, it serves as a food source for micro-organisms such as algae and bacteria. As a result, the population of these micro-organisms in the water body increases. Here, they utilize most of the dissolved oxygen for their metabolism. This results in an increase in the levels of Biological oxygen demand (BOD) in river water and results in the death of aquatic organisms. Also, the nutrients in the water lead to the growth of planktonic algal, causing algal bloom. This causes deterioration of water quality and fish mortality.

Q5. What is acid rain? What are the causes and effects of acid rain? How can the problem be overcome?

Acid rain is a form of precipitation characterised by high concentrations of acidic compounds, usually in the form of dissolved ions or molecules. Acid rain contains sulphur dioxide (SO_2) and nitrogen oxides (NOx) which are emitted from fossil fuel combustion. Acidic pollutants can also be discharged from the smokestacks of ships and coal-fired power plants.

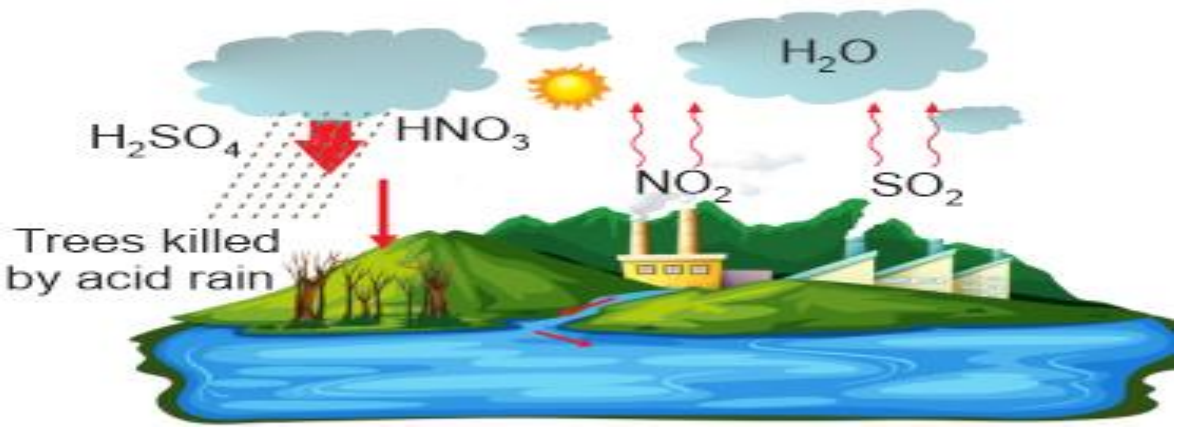
Acid rain is caused by a chemical reaction that begins when compounds such as sulphur dioxide and oxides of nitrogen are released into the air. These substances can rise very high up into the atmosphere, where they mix and react with water, oxygen, and other chemicals to form more acidic pollutants called acid rain. If the pH of rain water is less than 5.6, due to presence of pollutants gases known as acid rain. In acid rain primary pollutant react with moisture in the cloud to form secondary pollutants and fall as rain.

Causes of Acid Rain
Human Activities
The primary causes of acid rain are sulphur dioxide (SO_2) emitted from coal-fired power plants. Sulphur dioxide combines with oxygen to form sulphuric acid (H_2SO_4), which dissolves with the water droplets and rains down as acid rain. Nitrogen oxides emitted by automobiles also contribute to the formation of acid rain.

Natural Causes
Natural sources of acid rain include:
Volcanic eruptions – When volcanoes erupt, they release sulphur dioxide, which mixes with water vapour in the atmosphere to form sulphuric acid. It then reacts with rainwater and releases (H^+), which causes the water to turn acidic.

- **Seaspray –** When waves crash against rocks or the shoreline, they release aerosols into the atmosphere. These aerosols also decrease the pH level of rainwater.
- **Lightning** – Lightning is one of the most common natural sources of acid rain. This is because when it strikes, it creates an electrical discharge that travels from the ground to the air. This process creates nitrogen oxides (NOx) and sulphur dioxide (SO_2) gases that are released into the atmosphere.
- **Wildfires** – Wildfires are a natural part of the ecosystem, but they can cause acid rain by releasing sulphur dioxide into the air. When wildfires occur, they release large amounts of carbon dioxide and methane as well. These greenhouse gases can cause global warming, which leads to more wildfires.

Other sources include industrial emissions and biomass burning. Industrial emissions like chlorofluorocarbons (CFC), hydrochlorofluorocarbons (HCFCs) and hydrofluorocarbons (HFCs) are the main contributors. These chemicals deplete the ozone layer in the stratosphere, which absorbs ultraviolet radiation from the sun before reaching Earth's surface. Biomass burning includes forest fires, agricultural fires, and fossil fuels for heating or cooking purposes.



Which Gases are Responsible for Acid Rain?
The gases responsible for acid rain are carbon dioxide, nitrous oxide, and fluorinated gases such as hydrofluorocarbons (HFCs) and perfluorocarbons (PFCs). Acid rain reaction with these gases has been explained below.
Carbon Dioxide
$CO_2 + H_2O \rightarrow H_2CO_3$
$H_2CO_3 \rightarrow H^+ + HCO3^-$
Nitric Oxide
$NO_2 + H_2O \rightarrow HNO_3 + NO$
Sulphur Dioxide
$SO_2 + O_2 \rightarrow SO_3 + H_2O \rightarrow H_2SO_4$

Acid Rain — Effects
Acid rain is a local environmental problem that can have a global effect. Acid rain effects may take years to show up, and it is difficult to determine the cause of various problems. It can negatively affect human health and wildlife and damage buildings and property. Some of these effects include the following.

Acidification of Lakes and Streams
The pH level of most water bodies is around 6-8, however, it can drop below 4 when exposed to high concentrations of acidic substances like sulphuric acid or nitric acid from coal-fired power plants or factories. Acidic water can kill fish eggs by burning through their shells or making them too weak.

Damage to Buildings and Monuments
Acid rain can erode stone statues, buildings, bridges, and other structures made of limestone or marble. It can also damage buildings made of brick or concrete.

Reduced Visibility
The particles in acid rain can reduce visibility by hindering our view of the sun on a bright sunny day, causing squinting or even temporary blindness if we are standing too close to a source when acid rain occurs.

Health Effects
Acidic deposition can harm human health by damaging the lungs and causing respiratory problems like asthma attacks. It may also cause irritation in the eyes, such as conjunctivitis (pink eye), in people who exercise or play outdoor sports.

Acid rain affects humans indirectly. The sulphur dioxide and nitrogen dioxide created in the atmosphere can cause respiratory disorders or worsen these conditions. For example, people with respiratory conditions like asthma or chronic bronchitis might suffer from difficulty breathing.

Damage to Plants, Trees, and other Vegetation
Plants cannot thrive in an acidic environment because they require alkaline soil to grow properly. Acidic soil causes minerals such as calcium and magnesium to become less available for plant growth. This affects everything from root growth to flower development.

Oxidation or Rusting of Iron and Steel
Iron or steel come in contact with oxygen in acidic air and rust. This process is called oxidation or rusting. This is another effect of acid rain.

Prevention of Acid Rain
Now that you know the acid rain causes and effects, you should learn about the prevention of acid rain. The major acid rain prevention methods are:

- **Use of renewable energy:** Reducing fossil fuels by relying on renewable energy sources such as solar energy, wind energy, hydropower, etc.
- **Energy conservation:** Purchase energy-efficient electrical appliances and use public transportation, bicycles, or walking (especially short distances).
- **Liming:** Even if acid deposition has taken place, adding lime to water bodies and soil can help neutralize the pH.
- **Planting trees:** Trees absorb carbon dioxide from the air and release oxygen back into it again. This helps to balance out the effects of fossil fuels on our atmosphere.
- **Use public transport more often**: Reduce vehicle emissions by driving less often or using public transportation instead of driving alone.

Q6. Write the solution of paddy straw burning. Also write the adverse effects on the environment of paddy straw burning.
Solution of Paddy straw/stubble burning
Paddy straw burning can be controlled in the following ways. Some of the ways are listed below:
- The Government should introduce effective and economically viable technologies and infrastructure, such as the "Happy Seeder," to assist farmers in managing crop residue.

- Farmers should be encouraged to diversify their crops, moving away from paddy and towards fruits, cotton, vegetables, and maize.
- Local awareness programs should be organized to educate farmers about the harmful effects of stubble burning and provide them with viable alternatives.
- Agricultural scientist M.S. Swaminathan proposed the establishment of 'Rice Bio parks' in Haryana, Uttar Pradesh, and Delhi, which can convert stubble into useful products like cardboard, paper, and animal feed.
- Instead of burning stubble, the government should incentivize the conversion of crop residue into organic fertilizer, fodder, or fuel. The government should support industries involved in converting stubble into economically viable products.

Harmful/Adverse effects of paddy straw/stubble burning

The ill effects of paddy straw/stubble burning are listed below:

- Burning stubble can have an ill effect as it can result in the loss of wealth from the stubble.
- Burning of Stubble can increase pests or termites, as many microorganisms in the air are killed during burning. The Killing of this microorganism can lead to an increase in pests which can cause disease in crops.
- Every year when the stubble is burnt, nitrogen, sulfur, Phosphorus, organic carbon, and potassium are destroyed. This can otherwise be used to make Organic manure. It will further reduce the dependency and the use of chemical fertilizers.
- When the husk is burnt on the ground, it destroys or reduces the nutrients in the soil and makes it less fertile.
- The heat is generated after the burning of stubble or crop residue penetrates the soil, which results in the loss of moisture and useful microbes.
- Stubble burning is one of the major causes of air pollution in parts of North India, which deteriorates the air quality. The stubble burning contributes to the winter haze in Delhi and the winter smog. Stubble burning radiates many toxic pollutants into the atmosphere, including harmful gases like carbon monoxide, Methane, volatile organic compounds, and carcinogenic polycyclic aromatic hydrocarbons.
- Burning stubble for crown rot and take-all does not remove below-ground inoculum and can worsen crown rot's impact due to drier soil later in the season.
- Retaining stubble can aid in disease management; for instance, standing cereal stubble has been found to reduce ascochyta blight infection in chickpea crops, and similar benefits have been observed with lupin diseases sown into mulched cereal stubble.

Q7. What do you mean by the concept of Indian standard of automobile emission? Also explain the difference between BS4 and BS6 vehicles.

- Emission norms were introduced in India for petrol (gasoline)-powered vehicles in 1991 and for diesel-powered vehicles in 1992. Subsequently, India made the catalytic convertor mandatory for petrol vehicles and introduced unleaded petrol.
- The country's Supreme Court ruled that all vehicles in India must meet Euro I or India 2000 norms by June 1999; and that Euro II will be mandatory in New Delhi by April 2000. Manufacturers were unprepared for this upgrade and, in a subsequent ruling, the implementation date for Euro II was not enforced.
- In 2002, the Mashelkar committee recommended a phased implementation of future norms with the regulations being implemented in major cities first and later in the rest of the country.
- National Auto Fuel policy was announced in 2003. The roadmap for implementation of the Bharat Stage norms were not laid out until 2010. The policy also set guidelines for auto fuels; reduction of pollution from older vehicles; R&D for air quality data; and health administration.

The Central Pollution Control Board under the Ministry of Environment & Forests and Climate Change sets the standards and timeline for implementation. BS-III norms have been enforced across the country. In 13 major cities, BS-IV emission norms have been in place since April 2010. In 2016, the government announced that the country would skip the BS-V norms altogether and adopt BS-VI norms by 2020.

About BS-VI Norms

Bharat stage (BS) emission standards are laid down by the government to regulate the output of air pollutants from internal combustion engine and spark-ignition engine equipment, including motor vehicles.

India has adopted BS Emission Standards since 2000, modelled on European Union norms.
The first emission norms with the name ‘India 2000’ were introduced in the year 2000. BS2 and BS3 were introduced in 2005 and 2010, while BS4 norms came into effect in 2017 with stricter emission standards or norms. The BS standards regulate tailpipe emissions of air pollutants, including particulate matter, SOx, and NOx, as well as carbon monoxide, hydrocarbons, and methane.

Who sets them?

The standards and timeline for implementation are set by the Central Pollution Control Board under the Ministry of Environment, Forests and Climate Change.
In April 2020, India leapfrogged from BS-IV to the implementation of BS-VI. The central government has mandated that vehicle makers must manufacture, sell, and register only BS-VI (BS6) vehicles from April 1, 2020. It is applicable for cars, trucks, buses, three-wheelers, and two-wheelers (motorcycles, scooters, and mopeds). This emission norm does not apply to off-highway equipment and vehicles such as tractors, back-hoe loaders, excavators, etc.

Indian Emission Standards (4-Wheel Vehicles)

Standard	Reference	YEAR	Region
India 2000	Euro 1	2000	Nationwide
Bharat Stage II	Euro 2	2001	NCR*, Mumbai, Kolkata, Chennai
		2003.04	NCR*, 13 Cities†
		2005.04	Nationwide
Bharat Stage III	Euro 3	2005.04	NCR*, 13 Cities†
		2010.04	Nationwide
Bharat Stage IV	Euro 4	2010.04	NCR*, 13 Cities†
Bharat Stage V	Euro 5	(to be skipped)	
Bharat Stage VI	Euro 6	2020.04 (proposed)	Entire country

* National Capital Region (Delhi)
† Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad, Pune, Surat, Kanpur, Lucknow, Sholapur, Jamshedpur and Agra

“Emission Standards for four wheel vehicles in India”

Difference between BS-IV and BS-VI Vehicles:

Both BS-IV and BS-VI are unit emission norms that set the maximum permissible levels for pollutants that an automotive or a two-wheeler exhaust can emit.
Compared to BS4, BS6 emission standards are stricter.
The main difference is in the amount of sulphur in the fuel, which is reduced from 50 ppm in BS IV fuel to 10 ppm in BS VI fuel for both gasoline and diesel. Hence, BSVI engines produce less harmful emissions and pollutants.

Engine Type	Mass of exhaust gas	BS4 Limit	BS6 Limit	Percentage Decrease
PETROL	CO (in mg/km)	1000	1000	Nil
	HC (in mg/km)	100	100	Nil
	Nox (in mg/km)	80	60	2.50%
	PM (in mg/km)	.	4.5	
Diesel	CO (in mg/km)	500	500	Nil
	HC+Nox (in mg/km)	300	170	43%
	Nox (in mg/km)	250	80	68%
	PM (in mg/km)	25	4.5	82%

Q8. (i) What is El-Nino? What are its effects?

El Nino can be understood as a natural phenomenon wherein the ocean temperatures rise especially in parts of the Pacific Ocean. It is the nomenclature which is referred to for a periodic development along the coast of Peru. This development is a temporary replacement of the cold current along the coast of Peru. During El Niño, trade winds weaken. Warm water is pushed back east, toward the west coast of the Americas. El Niño typically peaks around December. El Niño can affect our weather significantly. The warmer waters cause the Pacific jet stream to move south of its neutral position. With this shift, areas in the northern U.S. and Canada are dryer and warmer than usual. But in the U.S. Gulf Coast and Southeast, these periods are wetter than usual and have increased flooding.

El Nino Effects

1. El Nino results in the rise of sea surface temperatures
2. It also weakens the trade winds of the affected region.
3. In India, Australia, it can bring about drought conditions. This affects crop productivity largely. It has been also observed certain times, that EL Nino may not bring drought but cause heavy rainfall. In both the cases, it causes heavy damage.
4. However, in some other countries, it may result in a complete reversal, i.e., excessive rainfall.

Impact on Indian Agriculture:

5. **Weak Monsoon for India:** The formation of an El Nino in May or June 2023 may result in a weakening of the southwest monsoon season, which accounts for roughly 70% of total rainfall in India and on which most farmers still rely.
6. **Sporadic increase in rainfall:** Sub-seasonal influences, such as the Madden-Julian Oscillation (MJO) and monsoon low-pressure systems, can temporarily increase rainfall in some areas, as shown in 2015.
7. **Drought:** Typically, El Nino causes drought. El Nino has previously produced drought or a weak southwest monsoon in the country.
8. A bad monsoon could have an impact on the growth of Kharif crops. Paddy, a water guzzler, groundnuts, and pulses are the crops most likely to suffer from a severe El Nino. Cotton and sugarcane crops may suffer as well.

Mitigation Of Effects:

1. Keeping a check on the sea surface temperatures
2. Maintaining sufficient buffer stocks of food grains and ensuring their smooth supply
3. Ensuring relevant support to the farmer community including economic help
4. Alternative ways to be promoted such as the practice of sustainable agriculture

(ii) 'The Ecological impacts of acid rains is quite serious'. Comment.

Acid Rain Harms Forests:

Acid rain can be extremely harmful to forests. Acid rain that seeps into the ground can dissolve nutrients, such as magnesium and calcium, that trees need to be healthy. Acid rain also causes aluminum to be released into the soil, which makes it difficult for trees to take up water. Trees that are located in mountainous regions at higher elevations, such as spruce or fir trees, are at greater risk because they are exposed to acidic clouds and fog, which contain greater amounts of acid than rain or snow. The acidic clouds and fog strip important nutrients from their leaves and needles. This loss of nutrients makes it easier for infections, insects, and cold weather to damage trees and forests.

Acid Rain Damages Lakes and Streams:

Without pollution or acid rain, most lakes and streams would have a pH level near 6.5. Acid rain, however, has caused many lakes and streams in the northeast United States and certain other places to have much lower pH levels. In addition, aluminum that is released into the soil eventually ends up in lakes and streams. Unfortunately, this increase in acidity and aluminum levels can be deadly to aquatic wildlife, including phytoplankton, mayflies, rainbow trout, small mouth bass, frogs, spotted salamanders, crayfish, and other creatures that are part of the food web. As a result, the entire food web is affected. So, although acid rain may not directly affect a certain species of plant or animal, it can affect the entire food web by limiting the amount of food available.

This problem can become much worse during heavy downpours of rain or when the snow begins to melt in the spring. These types of events are known as "**episodic acidification**".

Q9. At the onset of winter every year, stubble burning is invariably blamed for smog in and around Delhi; Why? What are the other factors contributing to smog?

Delhi witnesses a smoke-filled weather. As per the Indian Agricultural Research Institute (IARI), around 14 million tons (Mt) out of the 22 Mt of the rice stubble generated every year in India, is set to fire. This accounts to about 63.6 per cent of the generated rice stubble and Haryana and Punjab alone contribute 48 per cent to the amount of stubble being burnt.

Stubble burning can be defined as the intentional incineration of stubbles by farmers after crop harvest, as per a 2020 publicly available research study. Stubbles are essentially the cut stalks that are left on the field after harvesting the grains of cereal plants or stems of sugarcane. On a global level, stubble burning contributes to about one-fourth of the total biomass burning. While several factors contribute to the air pollution in the national capital, stubble burning in Delhi's neighboring states like Punjab, Haryana and Uttar Pradesh is considered to be a major cause behind it.

Other factors contributing to smog:

Air pollution in Delhi and NCR is a collective result of multiple factors including high level of anthropogenic activities in the high-density populated areas in NCR, arising from various sectors viz. Vehicular Pollution, Industrial Pollution, Dust from Construction and Demolition activities, Road and Open Areas Dust, Biomass Burning, Municipal Solid Waste burning, Fires in Landfills and air pollution from dispersed sources, etc. During post-monsoon and winter months, lower temperature, lower mixing heights, inversion conditions and stagnant winds lead to trapping of the pollutants resulting in high pollution in the region.

What is the solution?

Countries across South Asia will have to coordinate efforts if the region's pollution problem is to be solved, collaborating to enhance monitoring and make policy decisions. At the same time, these region-wide efforts will have to be balanced by moulding solutions to suit local conditions where needed. In addition, the focus will also have to be broadened to include sectors that have received limited attention so far, such as agriculture and waste management.

To curb stubble burning, for example, governments can offer subsidies on better harvesting machines. Countries like India have already started offering such incentives but demand for such machines has been limited due to their high purchase cost and high waiting time for those who want to rent them.

Q 10. What is Climate change? Write the causes and effects of climate change?

Climate change

Climate change refers to long-term changes in temperatures and weather patterns. These changes may be due to natural causes, but since the 19th century, new forms of production and consumption have contributed significantly to global warming. This fact poses a great challenge to the future of the planet, but we still have an opportunity to slow down this process.

The main cause of climate change is the greenhouse effect, a natural phenomenon that, under normal conditions, far from being harmful it's what allows our planet to maintain the conditions needed to support life. Certain gases in the atmosphere retain part of the thermal radiation emitted by the Earth’s surface after being heated by the sun, keeping the temperature at a level suitable for life to develop. The greenhouse effect is necessary. Without it, the Earth's average temperature would be around -15°C.

Natural Causes

Strength of the Sun

Almost all of the energy that affects the climate on Earth originates from the Sun. The Sun's energy passes through space until it hits the Earth's atmosphere. Only some of the solar energy intercepted at the top of the atmosphere passes through to the Earth's surface; some of it is reflected back into space and some is absorbed by the atmosphere. The energy output of the Sun is not constant: it varies over time and this has an impact on our climate.

Changes in the earth's orbit, axial tilt and precession

The three changes in the Earth's orbit around the Sun — eccentricity, axial tilt, and precession —affects the amount of solar heat that reaches the Earth's surface and subsequently influences climatic patterns, including periods of glaciation (ice ages).

Changes in ocean currents

Ocean currents carry heat around the Earth. As the oceans absorb more heat from the atmosphere, sea surface temperature increases and the ocean circulation patterns that transport warm and cold water around the globe change. The direction of these currents can shift so that different areas become warmer or cooler. As oceans store a large amount of heat, even small changes in ocean currents can have a large effect on global climate. In particular, increases in sea surface temperature can increase the amount of atmospheric water vapour over the oceans, increasing the quantity of greenhouse gas. If the oceans are warmer they can't absorb as much carbon dioxide from the atmosphere.

Plate tectonics

Plate tectonic processes cause continents to move to different positions on the Earth. The movement of the plates also causes volcanoes and mountains to form and these can also contribute to a change in climate. Large mountain chains can influence the circulation of air around the globe, and consequently influence the climate. For example, warm air may be deflected to cooler regions by mountains.

volcanic eruptions

Volcanoes affect the climate through the gases and particles (tephra/ash) thrown into the atmosphere during eruptions. The effect of volcanic gases and dust may warm or cool the Earth's surface, depending on how sunlight interacts with the volcanic material. During major explosive volcanic eruptions, large amounts of volcanic gas, aerosol droplets and ash are released. Ash falls rapidly, over periods of days and weeks, and has little long-term impact on climate change. In the present day, the contribution of volcanic emissions of CO_2 into the atmosphere is very small; equivalent to about one per cent of anthropogenic (caused by humans) emissions.

Changes in land cover

On a global scale, patterns of vegetation and climate are closely correlated. Vegetation absorbs CO_2 and this can buffer some of the effects of global warming. On the other hand, desertification amplifies global warming through the release of CO_2 because of the decrease in vegetation cover.A decrease in vegetation cover, via deforestation for example, tends to increase local albedo, leading to surface cooling.

Man-made causes

Agriculture

There are many significant ways in which agriculture impacts climate change. From deforestation in places like the Amazon to the transportation and livestock that it takes to support agricultural efforts around the world, agriculture is responsible for a significant portion of the world's greenhouse gas emissions. However, agriculture is also an area that is making tremendous strides to become more sustainable. As productivity increases, less carbon is being emitted to produce more food. Agriculture also has the potential to act as a carbon sink and could eventually absorb nearly the same amount of CO_2 it emits.

Deforestation

Deforestation and climate change often go hand in hand. Deforestation is the second leading contributor to global greenhouse gasses. Many people and organizations fighting against climate change it is one of the most important issues that must be addressed to slow or prevent climate change.

Human Activity

Burning of fossil fuels for electricity, heat, and transportation is another reason of climate change. Transportation in the form of cars, trucks, ships, trains, and planes emits the largest percentage of CO_2–speeding up global warming and remaining a significant cause of climate change.

Immediate Effects of Climate Change

From melting glaciers to more extreme weather patterns, people everywhere are beginning to take notice of the real impacts of climate change. While some nations around the world are taking action with initiatives such as the Paris Climate Agreement, others are continuing business as usual–pumping millions of tons of carbon into the atmosphere year after year. While the long-term consequences are still to be seen, for now, climate change continues to cause extreme weather as well as safety and economic challenges on a global scale.

Extreme Weather

Changes to weather are perhaps the most noticeable effect of climate change for the average person. Extreme weather influenced by climate change includes:

- Stronger storms & hurricanes
- Heatwaves
- Wildfires
- More flooding
- Heavier droughts

Safety & Economic Challenges

Long Term Impact of Climate Change

The long term impact of climate change could be absolutely devastating to the planet and everyone and everything living on it. If the world continues on its current trajectory, then we will likely continue to see increasing effects on everyday life.

Health

Climate change is already affecting the health of many and has the potential to impact millions more. According to the Center for Disease Control and Prevention, climate change-related health risks may include:

- Heat-related illness
- Injuries and fatalities from severe weather
- Asthma & cardiovascular disease from air pollution
- Respiratory problems from increased allergens
- Diseases from poor water quality
- Water & food supply insecurities

Negative Impact in Ecosystems

Ecosystems are interconnected webs of living organisms that help support all kinds of plant and biological life. Climate change is already changing seasonal weather patterns and disrupting food distribution for plants and animals throughout the world, potentially causing mass extinction events. Some studies estimate that nearly 30% of plant and animal species are at risk of extinction if global temperatures continue to rise.

Water & Food Resources

Climate change could have a significant impact on food and water supplies. Severe weather and increased temperatures will continue to limit crop productivity and increase the demand for water. With food demand expected to increase by nearly 70% by 2050, the problem will likely only get worse.

Sea Levels Rising
Rising sea levels could have far-reaching effects on coastal cities and habitats. Increasing ocean temperatures and melting ice sheets have steadily contributed to the rise of sea levels on a global scale.
Shrinking Ice Sheets
While contributing to rising sea levels, shrinking ice sheets present their own set of unique problems, including increased global temperatures and greenhouse gas emissions. Climate change has driven summer melt of the ice sheets covering Greenland and Antarctica to increase by nearly 30% since 1979.

Ocean Acidification
The ocean is one of the main ways in which CO_2 gets absorbed. While at first glance that may sound like a net positive, the increasingly human-caused CO_2 is pushing the world's oceans to their limits and causing increased acidity. As pH levels in the ocean decrease, shellfish have difficulty reproducing, and much of the oceans' food cycle becomes disrupted.

Q11. What is population growth? Explain various processes which determine population growth.

Population growth is one of the major concerns of the present world as the human population is not a static factor. Rather, it is growing at a very alarming rate. In spite of the increasing world population, the resources of the earth remain constant. Thus, the ability to maintain sustainable development is becoming a major challenge to mankind today.

Population growth is the increase in the number of people in a population or dispersed group.
Consequently, global human population rapidly increased, and continues to do so, with dramatic impacts on global climate and ecosystems. We will need technological and social innovation to help us support the world's population as we adapt to and mitigate climate and environmental changes.

The fluctuations in the population in a given area are influenced by four major factors, which include the following:

- **Natality –** It is the number of births in a given period of time in a population
- **Mortality** – It is defined as the number of deaths that takes place in a population at a given period of time.
- **Immigration** – It is defined as the number of individuals who come from another population and add to the population under consideration during a period of time.
- **Emigration** – It is defined as the number of individuals from a population who leave the habitat and go to a different habitat at a given period of time.

Thus, it is clearly visible, that Natality (N) and Immigration (I) add to a population, thus increasing the population whereas, Mortality (M) and Emigration (E) decrease the population. The population density (P_t) at a given point of time can be given as:

$$\mathbf{P_t = P_0 + (N + I) – (M + E)}$$

where P_0 is the initial population density.

We have two growth models which describe the basic growth trend in a population. These are:

1. **Exponential growth –** In an ideal condition where there is an unlimited supply of food and resources, the population growth will follow an exponential order. Consider a population of size N and birth rate be represented as *b*, death rate as d, the rate of change of N can be given by the equation

$$dN/dt = (b-d) \times N$$

$$\text{If, } (b – d) = r,$$

$$dN/dt = rN$$

Where r = intrinsic rate of natural increase

This equation can be represented with a graph which has a J shaped curve. According to calculus

$$N_t = N_0 e^{rt}$$

Where, N_t = Population density at time t

N_0= Population density at time zero

r = intrinsic rate of natural increase

e = base of natural logarithms

t = time

2. **Logistic growth** – This model defines the concept of 'survival of the fittest'. Thus, it considers the fact that resources in nature are exhaustible. The term 'Carrying capacity' defines the limit of the resources beyond which they cannot support any number of organisms. Let this carrying capacity be represented as K.

The availability of limited resources cannot show exponential growth. As a result, the graph will have a lag phase, followed by an exponential phase, then a declining phase and ultimately an asymptote. This is known as Verhulst-Pearl Logistic Growth and is represented using the equation:

$$dN/dt = rN((K-N)/K)$$

Q12. How the population of any country can be analyzed?

The population of a country is calculated using several methods, primarily through censuses, surveys, and statistical estimates. Here's an overview of the main methods:

1. **Census**:
   - A census is a comprehensive count of the population conducted at regular intervals (usually every 10 years). It collects demographic data, including age, sex, ethnicity, and housing information.
   - In many countries, a national census is mandated by law, and it involves enumerators visiting households to gather information.
2. **Surveys**:
   - National surveys, such as the American Community Survey in the United States, are conducted in between censuses to gather demographic and economic data. These surveys help update population estimates and provide insights into specific population segments.
3. **Vital Statistics**:
   - This method involves tracking births, deaths, and migration. Vital registration systems record these events, which are crucial for understanding population changes.
   - Birth and death rates are used to calculate natural population increase or decrease.
4. **Administrative Records**:
   - Government agencies maintain records that can be used to estimate population size. These include tax records, social security registrations, and school enrollment data.
   - Some countries use these records to create estimates without conducting a full census.
5. **Statistical Models**:
   - Demographers often use statistical models to estimate population figures, especially in countries where data may be incomplete or unreliable. These models can incorporate trends from previous censuses, surveys, and vital statistics.
6. **Migration Estimates**:
   - Migration significantly affects population size. Estimates of immigration and emigration are often based on surveys, border control data, and other administrative records.
7. **Population Projections**:
   - Using the data collected from the above methods, demographers create projections to estimate future population growth or decline based on current trends.

The combination of these methods allows for a comprehensive understanding of a country's population dynamics. Each country may have its own specific procedures and frequency for conducting these methods, leading to variations in data collection and reporting.

Unit 5

Ans1 Two Objectives of Environmental Education:

1. **Awareness Creation:** To create awareness among individuals about environmental issues and the importance of sustainable living.
2. **Behavioral Change:** To encourage individuals to adopt environmentally friendly practices and contribute to conservation efforts.

Ans2. Environmental Ethics:

Environmental ethics refers to the moral relationship between humans and the natural environment. It involves understanding the ethical implications of human actions on ecosystems and the planet, advocating for sustainable practices, and ensuring the protection of biodiversity and natural resources for future generations.

Ans3. Montreal Protocol:

The **Montreal Protocol** is an international treaty adopted in 1987 aimed at protecting the ozone layer by phasing out the production and use of ozone-depleting substances (ODS). It is one of the most successful environmental agreements, as it has significantly reduced the use of chemicals like CFCs (chlorofluorocarbons), contributing to the recovery of the ozone layer.

Ans4. Two Objectives of Environmental Education (Repeated):

1. To raise awareness about environmental issues.
2. To promote sustainable practices and actions among individuals and communities.

Ans5. Population Density:

Population density refers to the number of individuals living per unit of area, typically per square kilometer or square mile. It gives an indication of how crowded a specific area is.

Ans6. Logistic Growth of Population:

Logistic growth describes a population's growth when resources are limited. Unlike exponential growth, where the population increases rapidly, logistic growth considers environmental constraints and leads to a stabilization of the population size when it reaches the carrying capacity of its environment.

Ans7. Kyoto Protocol:

The **Kyoto Protocol** is an international agreement adopted in 1997 that commits industrialized countries to reduce their greenhouse gas emissions to combat global warming and climate change. It sets legally binding targets for these countries to lower emissions by an average of 5.2% below 1990 levels by the year 2012.

Ans8. Doubling Time:

Doubling time refers to the period required for a population or quantity to double in size or value, assuming it grows at a constant rate. This concept is often used in population studies to predict how long it will take for a population to double given its current growth rate.

Long Questions

Ans1. National NGOs Working for Environmental Protection in India:

- **Centre for Science and Environment (CSE):** Focuses on sustainable development through research, advocacy, and policy influence. Known for its work on air/water pollution and climate change.
- **WWF-India:** Works on wildlife conservation (e.g., tigers, elephants), habitat restoration, and climate action.
- **The Energy and Resources Institute (TERI):** Promotes clean energy, waste management, and sustainable agriculture.
- **Vanashakti:** Engages in forest conservation, mangrove protection, and environmental litigation.
- **Greenpeace India:** Campaigns against deforestation, plastic pollution, and coal reliance through grassroots activism.

Ans 2. International NGOs for Environmental Protection:

- **Greenpeace International:** Global campaigns against deforestation, fossil fuels, and ocean plastic.
- **World Wildlife Fund (WWF):** Focuses on biodiversity conservation (e.g., pandas, polar bears) and sustainable practices.
- **Friends of the Earth (FoE):** Advocates for climate justice, anti-plastic initiatives, and corporate accountability.
- **IUCN (International Union for Conservation of Nature):** Manages the Red List of Threatened Species and promotes ecosystem restoration.
- **350.org:** Mobilizes global movements to phase out fossil fuels and promote renewable energy.

Ans3. Schemes Promoting Women’s Education in India:

- **Beti Bachao Beti Padhao (BBBP):** Addresses gender bias and promotes girls' enrollment in schools.
- **Sukanya Samriddhi Yojana (SSY):** Savings scheme for girls’ education and marriage expenses.
- **National Scheme of Incentive to Girls for Secondary Education:** Provides cash transfers for girls completing secondary school.
- **CBSE Udaan:** Free online resources and mentoring for girls in STEM.
- **Kasturba Gandhi Balika Vidyalaya (KGBV):** Residential schools for girls from marginalized communities.

Ans 4. Obstacles in Women’s Education:

- **Cultural Norms:** Preference for boys’ education and early marriage.
- **Economic Barriers:** Poverty forces girls into labor instead of school.
- **Safety Concerns:** Lack of safe transportation and harassment.
- **Infrastructure Gaps:** Absence of separate toilets and female teachers.
- **Gender Stereotypes:** Belief that education is unnecessary for girls.

- **Gap in upper primary and secondary schooling:** While female enrolment has increased rapidly since the 1990s, there is still a substantial gap in upper primary and secondary schooling.
- **High drop-out rates:** Increased female enrolment is, compromised by persistently high rates of drop-out and poor attendance of girls relative to boys. Girls also constitute a large proportion of out-of-school children.
- **Inter-state variations:** There are also considerable inter-state variations in gender parity. While the greatest surges in female enrolment have been achieved in the most educationally disadvantaged states such as Bihar and Rajasthan, these states still have a long way to go to catch up with the better performing states of Kerala, Tamil Nadu and Himachal Pradesh.
- **Quality of education and infrastructure:** Within government schools- overcrowded classrooms, absent teacher, unsanitary conditions, absence of girl’s toilet are common complaints and can cause parents to decide that it is not worth their girl child going to school.
- **Social factors:** Early marriages as per their social custom, girl children are not allowed to go outside the house and village because it is a social taboo, parents go to their workplaces and household activities are undertaken by the young female children, caring of younger ones at home, gender disparity at home, in society and earlier marriages in this region.
- **Health factors:** frequent ill-health of the student especially female due to lack of nutritious food and unhygienic conditions in living areas.
- **Economic factors:** Due to difficult financial conditions and social conditioning, families usually neglect girl education because which they are not employment ready.
- **Patriarchy and social Perceptions:** Position of women in the social, political and economic system, is very low. They are noticeably absent from the discussions of development theory too. This leads to absence of a dominant voice for assertion of their needs and rights as a citizen.
- **Son preference:** Some studies suggest that girls are over-represented in the government schools, demonstrating continuing son preference where boys (highlighted in economic survey 2018) are educated in private and better schools which are of (perceived) better quality.

Ans5. Salient Features of the Environmental Protection Act (EPA), 1986 (India):

- The act empowers the central Government with all the power to take all such measures as it deems necessary or expedient for the purpose of:
  - protecting and improving the quality of the environment and

    - preventing controlling and abating **environmental pollution**, which includes setting new national standards for the quality of the environment (ambient standards) as well as standards for controlling emissions and effluent discharges;
    - regulating industrial locations,
    - prescribing procedures for managing hazardous substances;
    - establishing safeguards to prevent accidents and collecting and dismantling information regarding environmental pollution.
- Under the act, the Central Government was also empowered including the power of entry for examination, testing of equipment and other purposes and power to analyze the sample of air, water, soil or any other substance from any place.
- There is also a specific prohibition on handling hazardous substances except those in compliance with regulatory standards and procedures.
- The act also provides for penalties for violating the provisions of the act. Any person who fails to comply with the Act shall be punished with a prison term of up to five years or fine up to Rs. 1 lakh, or both.
- The act also has innovative provisions for its enforcement, which was not present in any other pollution control legislation at that time.
    - It provides that in addition to authorised Government officials, any person may file a complaint with a court, alleging an offence under the Act.
    - Every rule made under this Act shall be laid, as soon as may be after it is made, before each House of Parliament.

Ans6. Population Explosion in India:

Population Explosion refers the sudden and rapid rise in the size of population, especially human population. It is an unchecked growth of human population caused as a result of:

- increased birth rate,
- decreased infant mortality rate, and
- improved life expectancy.

The causes of population explosion are as follows:

- Accelerating birthrate: Due to lack of awareness about the positive impact of using birth-control method, there has been a steady growth in birthrate.
- Decrease in infant mortality rate: An improvement in medical science and technology, wide usage of preventive drugs (vaccines), has reduced the infant mortality rate. There has been great improvement in medical and health-care facilities during the past few decades.
- Increase in life expectancy: Due to improved living conditions, better hygiene and sanitation habits, better nutrition, health education, etc. the average life expectancy of human population has improved significantly. Steady supply of good quality food make sure that the population is well nourished. Populations grow when they are adequately nourished.
- Increased immigration: An increase in immigration often contributes towards population explosion, particularly in developed countries. It happens when a large number arrive at an already populated place with the intention to reside permanently.
- Less space than required: In urban cities, it is often found that there is very less scope for making available extra space to absorb the additional population. In such cases, a large population is seen packed into a smaller space.

- **Objectives of Sound Population Policy:**
    - Improve access to contraception and reproductive healthcare.
    - Promote women’s education and employment.
    - Implement awareness campaigns on family planning.
    - Address cultural norms through media and community engagement.
    -

Ans7. Characteristics of Population Growth:

- **Birth/Death Rates:** High birth rates in Nigeria vs. low in Japan.
- **Migration:** Urbanization in India due to rural-urban shifts.

- **Age Structure:** Youthful population in India vs. aging in Germany.

Population attributes are the features of the population that defines its characteristic like growth pattern, population density and, population regulation, these attributes are as follows,

1. Sex ratio
2. Mortality rate
3. Natality rate
4. Dispersion
5. Population density

Sex Ratio

At the individual level, the individual can identify as male or female, whereas in population biology or at the population level the quantitative measurements are called sex ratio. The sex ratio can be defined as the number of females per male, it is the ratio between the number of males the female. It is generally considered for sexually reproducing dioecious populations.

Mortality Rate

Mortality can be defined as the death of an individual in the population. The mortality rate is also commonly known as the death rate. The mortality rate can be defined as the number of individuals who died during a given period of time. The mortality rate is often calculated as the number of deaths per 1000 individuals per year. It is important to note that the mortality rate is time specific.

Natality Rate

The natality rate is also known as the birth rate. The term natality is more commonly used in population biology when describing a study of the human population. Natality is defined as the birth of an individual in a population, whereas the mentality rate refers to the number of individuals produced per female per unit of time. They are an important factor used in the calculations of the dynamics of the population.

Dispersion

Dispersion can be defined as the spatial and temporal distribution pattern of the individual of the population. There are two groups or classifications of the method of dispersion are known as random dispersion and clumped dispersion.

Population density

Population density refers to the **number of individuals living per unit area**, typically expressed as people per square kilometer or mile[1]. It helps understand how crowded or sparse a location is and provides insight into land use patterns, resource distribution, and environmental impact.

Ans 8. Factors Changing Population Pyramid Shape:

- Declining birth rates (e.g., China's one-child policy).
- Improved healthcare lowering death rates.
- Immigration influx (e.g., UAE's expatriate workforce).

9. Institutional & Societal Changes for Women's Empowerment:

- **Institutional:** Enforce equal pay laws, gender quotas in legislatures, and anti-discrimination policies.
- **Societal:** Challenge patriarchal norms, promote shared household responsibilities, and normalize women's leadership roles.

Ans 10. Population Explosion & Effects:

A rapid increase in the population is known as a population explosion.

The two causes of population explosion are:

- Increase in the birth rate.
- Decrease in the death rate.

An increase in population is directly related to the birth rate. The population in India is rising at a considerable speed, which results in various problems, including social, economic and environmental.

Increasing population has many adverse effects on our environment.

Listed below are the significant effects of increasing population:

- Increase in slum areas.
- Rise in unemployment and underemployment.
- Eradicating poverty and hunger becomes more difficult.
- Providing better health and education facilities becomes more difficult.
- This can lead to less availability of land, water, food and other necessary resources.
- An increase in population may lead to the unavailability of food grains due to the excessive demands of people.
- Overexploitation of water and other natural resources results in no scope of replenishment, which is a recipe for a natural disaster.
- Increasing population increases air and water pollution, leading to an increase in diseases, which ultimately leads to a rise in expenditure on healthcare.